

Dynamic modelling and analysis of human brucellosis on age heterogeneity in Jinzhou, China

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ABSTRACT

Since brucellosis causes harms and losses to sheep and human populations in the cities of Northern and Northwest China, we establish a cross-species brucellosis model with age-structure and bilinear incidences to characterize brucellosis transmission mechanism in this study. Firstly, the basic reproduction number, the disease-free and endemic equilibrium points of the brucellosis model are obtained. Then, the stabilities of two equilibrium points are proved by Routh-Hurwitz Criterion and LaSalle's Invariance Principle. Precisely, the disease-free equilibrium point is locally and globally asymptotically stable with $R_0 < 1$; and the endemic equilibrium point is globally asymptotically stable with $R_0 > 1$. Further, the 2017–2023 monthly brucellosis surveillance data from Jinzhou CDC are fitted by the least squares method and EpiSIX, showing that the values of basic reproduction numbers keep high consistency with the known results. The numerical simulations indicate that the reduction of brucellosis transmission risks depend on the decreases of direct transmission rates and recruitment rate, and the increases of culling rate and vaccination rate. Finally, the 2024–2030 yearly tendency predictions under two scenarios indicate that quadruple-control measure provides the extreme brucellosis transmission risks, compared with those from three-type control measures. As a consequence, for the local farmers and policy-makers, some suggestions are proposed to suppress the brucellosis infectious scales for the cities with similar sheep breeding patterns and geographical locations to Jinzhou on the Chinese mainland.

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1. Introduction

Brucellosis is one of the major zoonoses caused by various brucella species, and brucellosis is transmitted between animal population and human population through multiple routes, including contact with infected animals and their secretions, ingestion of contaminated water or animal products, and inhalation of aerosols (Rees et al., 2021). Brucellosis causes severe and persistent harms and losses to animal population and human population (World Health Organization, 2020; National Disease Control and Prevention Administration of the People's Republic of China, 2023). To animal population, brucellosis mainly affects reproductive function and fertility of adult animals, and reduces the survival rate of newborns (World Organisation for Animal Health, 2025). To human population, brucellosis rarely causes deaths but could persist if not treated in a timely manner (Pappas et al., 2006; Shandong Public Health Clinical Center, 2021). Across the globe, there exist approximately 500,000 new brucellosis cases every year. On the Chinese mainland, the brucellosis cases were reported in the early 1900s, increased from the 1950s to the 1970s, gradually reduced in the 1980s and the 1990s, and rebounded rapidly from 2000 to 2016. The recent work pointed out that brucellosis incidence on the Chinese mainland showed an upward trend again from 2017 to 2021 (Wang et al., 2023). Therefore, it is crucial to formulate targeted prevention and control measures for brucellosis epidemics to suppress the upward trend in China.

To explore the transmission mechanism and brucellosis transmission risks, the mathematical models are governed to compare the efficacy of distinct prevention and control measures. Some mathematical models in (Gong et al., 2023; He et al., 2025; Hou et al., 2013; Hou et al., 2014; Liu et al., 2025; Sun & Zhang, 2014; Zhang et al., 2024) without host population group or spatial distribution were investigated with the extensive explorations by different prevention and control measures. Sun and Zhang (2014) pointed out that the reduction of sheep mobility and the implementation of sheep self-breeding were key to control sheep brucellosis. Liu et al. (2025) further derived that the effective control of sheep population depended on the comprehensive implementation of vaccination, disinfection and culling infected sheep. Gong et al. (2023) in an empirical study investigated that the high-prevalence risk of Ningxia was controlled by increasing the culling rate and the vaccination rate of sheep population, and reducing the exposure risk of human population through health education. He et al. (2025) pointed out that the small values of two delays to sheep population and human population made the brucellosis infection scales less and the peak flattened, when the optimal prevention and control measures for brucellosis epidemics were taken into account. These investigations with prevention and control measures established the theoretical foundations for reducing brucellosis incidence, and also provided the important supports for the sustainable development of livestock farming and public health security. As a consequence, the above investigations seldom took the heterogeneity in groups and spatial distribution of sheep population into account, when describing the transmission mechanisms of sheep brucellosis within different regions.

To tackle the aspects mentioned above, the recent studies (Li et al., 2014, 2017, 2019; Qin et al., 2022; Yang et al., 2017, 2019, 2025; Zheng et al., 2025) investigated the multi-group mathematical models of sheep population. Li et al. (2017) established a multi-group SEIW brucellosis transmission model without human population to describe the variety of brucellosis transmission mechanism. Yang et al. (2017) considered the spatial transmission of brucellosis in a two-patch setting, and further found that the spatial mobility heterogeneity of sheep population as crucial for brucellosis transmission. Li et al. (2014) studied a multi-group dynamic model adapting the public farms in China, with two-way mixed cross-infection between cattle and sheep, and showed that no mixed feeding was an essential measure to eliminate brucellosis. Li et al. (2019) constructed a brucellosis dynamic model incorporating sheep population with partition of basic ewe population and other sheep population, and demonstrated that the enhancement of vaccination and seropositive detection rates of sheep population reduced brucellosis transmission risks. Qin et al. (2022) established a two-patch dynamic model by taking Shanxi and Hebei as case studies, and showed that brucellosis was effectively controlled by reducing sheep immigration from Hebei with low-infection to Shanxi with high-infection, and allowing moderate reverse immigration to accelerate this reduction. Yang et al. (2019) constructed a brucellosis dynamic model incorporating infection age and spatial diffusion characteristics of sheep population, and discovered that cleanliness of living environment and elimination of infectious sheep were effective measures. Zheng et al. (2025) constructed a brucellosis model incorporating age-structure by dividing susceptible sheep into lambs and adult sheep to explore the impacts of heterogeneity of age-structure on brucellosis transmission risks. The above investigations provided the impacts of heterogeneity in groups and spatial distribution of sheep population with the extensive explorations. But, the investigations on the heterogeneity of human population regarding transmission risks are rarely discussed in the recent works.

We develop a cross-species brucellosis model incorporating human age groups in Jinzhou to investigate the prevalence trend of brucellosis from the theoretical and practical perspectives in this study. The establishment of brucellosis model, the expression of basic reproduction number, and the existence of disease-free and endemic equilibrium points are discussed in Section 2. The local stability of the disease-free equilibrium point by Routh-Hurwitz Criterion is showed, and the global stabilities of both equilibrium points by LaSalle's Invariance Principle are explored in Section 3. The numerical simulations of Jinzhou brucellosis epidemics are performed by the least squares method and EpiSIX, and the values of basic reproduction numbers from the two methods are compared with the known results. Meanwhile, the sensitivity of main parameters to the peak of infected individuals is analyzed by the Partial Rank Correlation Coefficient method, and the prevalence trend of Jinzhou brucellosis epidemics is predicted in Section 4. The main conclusions and relevant suggestions are provided from the prevention and control perspectives of sheep population and human population in Section 5.

2. Model formulation

2.1. Formulation of cross-species epidemic model

In this study, we develop a cross-species epidemic model for describing brucellosis transmission mechanism. Within sheep population $N_s(t)$ at time t , the four compartments consisting of susceptible sheep $S_s(t)$, exposed sheep $E_s(t)$, infectious sheep $I_s(t)$, and vaccinated sheep $V_s(t)$ are taken into account. The brucellosis spreads among sheep population through two pathways: direct transmission from sheep contacts, and indirect transmission from brucella inhalation in the environment. $W(t)$ is the average number of brucella in the environment at time t that causes brucellosis infection. To characterize the transmission dynamics within sheep population and brucella in the environment, we have the following assumptions: (i) The direct transmission rates from exposed and infectious sheep to susceptible sheep are equal. (ii) The brucella shedding rates of exposed and infectious sheep are equal. Therefore, we establish sheep-brucella model (1) with the bilinear incidence below.

$$\begin{cases} \frac{dS_s(t)}{dt} = \Lambda_s - \beta_s S_s(E_s + I_s) - \gamma_s S_s W + \delta_s V_s - \sigma_s S_s - \mu_s S_s, \\ \frac{dE_s(t)}{dt} = \beta_s S_s(E_s + I_s) + \gamma_s S_s W - \theta_s E_s - \mu_s E_s, \\ \frac{dI_s(t)}{dt} = \theta_s E_s - \alpha_s I_s - \zeta_s I_s - \mu_s I_s, \\ \frac{dV_s(t)}{dt} = \sigma_s S_s - \delta_s V_s - \mu_s V_s, \\ \frac{dW(t)}{dt} = \kappa(E_s + I_s) - \tau W - \eta \epsilon W. \end{cases} \quad (1)$$

The meanings of the main parameters of sheep population are explained below. Λ_s is the constant recruitment rate; β_s is the direct transmission rate from exposed and infectious sheep to susceptible sheep; γ_s is the indirect transmission rate from brucella in the environment to susceptible sheep; σ_s is the vaccination rate; δ_s is the loss rate of vaccination immunity; μ_s is the natural mortality rate; θ_s is the incubation rate; α_s is the brucellosis-induced mortality rate; and ζ_s is the culling rate of infectious sheep. Then, the meanings of main parameters of brucella in the environment are explained. κ is the brucella shedding rate of exposed and infectious sheep; τ is the decaying rate; η is the number of disinfection times; and ϵ is the effective disinfection rate.

The interactions among human population, sheep population, and brucella in the environment are frequently encountered in reality. In this study, we would establish a human age-structured model to investigate how human age groups affect the transmission mechanism. First, we separate the human population $N_h(t)$ at time t into two age groups, including susceptible young individuals $S_{h_1}(t)$, infected young individuals $I_{h_1}(t)$, susceptible old individuals $S_{h_2}(t)$, and infected old individuals $I_{h_2}(t)$. We then notice that exposed sheep and infectious sheep and their carcasses trigger the infection risk. Especially, the infection risk from these carcasses can not be neglected in reality, because these carcasses with the persistent brucella are contacted by susceptible individuals. The losses of susceptible individuals originate from three aspects: the loss from exposed and infectious sheep, the loss from their carcasses, and the loss from brucella in the environment. For each age group of human population, we assume that the direct transmission rates from exposed sheep, infectious sheep, and their carcasses towards susceptible human population are equal. So, we adopt $\beta_{h_1} S_{h_1}(E_s + I_s)$ to characterize the loss from exposed and infectious sheep, and $\beta_{h_1} S_{h_1}(\mu_s E_s + (\alpha_s + \mu_s) I_s)$ to characterize the loss from their carcasses, and establish a human age-structured model (2) below.

$$\begin{cases} \frac{dS_{h_1}(t)}{dt} = \Lambda_h - \beta_{h_1} S_{h_1}((1 + \mu_s)E_s + (1 + \alpha_s + \mu_s)I_s) - \gamma_{h_1} S_{h_1} W - \mu_{h_1} S_{h_1} - \xi S_{h_1}, \\ \frac{dI_{h_1}(t)}{dt} = \beta_{h_1} S_{h_1}((1 + \mu_s)E_s + (1 + \alpha_s + \mu_s)I_s) + \gamma_{h_1} S_{h_1} W - \mu_{h_1} I_{h_1} - \xi I_{h_1}, \\ \frac{dS_{h_2}(t)}{dt} = \xi S_{h_1} - \beta_{h_2} S_{h_2}((1 + \mu_s)E_s + (1 + \alpha_s + \mu_s)I_s) - \gamma_{h_2} S_{h_2} W - \mu_{h_2} S_{h_2}, \\ \frac{dI_{h_2}(t)}{dt} = \xi I_{h_1} + \beta_{h_2} S_{h_2}((1 + \mu_s)E_s + (1 + \alpha_s + \mu_s)I_s) + \gamma_{h_2} S_{h_2} W - \mu_{h_2} I_{h_2}. \end{cases} \quad (2)$$

Here, Λ_h is the constant recruitment rate of human population; β_{h_1} and β_{h_2} are respectively the direct transmission rates from exposed and infectious sheep to susceptible humans with two age groups; γ_{h_1} and γ_{h_2} are respectively the indirect transmission rates from brucella in the environment to susceptible humans with two age groups; μ_{h_1} and μ_{h_2} are respectively the natural mortality rates of humans with two age groups; and ξ is the conversion rate from young group to old group. The brucellosis transmission mechanism within sheep population, brucella and human population is presented in models (1)–(2) and Fig. 1. The initial conditions of models (1)–(2) are given by $S_s(0) \geq 0, E_s(0) \geq 0, I_s(0) \geq 0, V_s(0) \geq 0, W(0) \geq 0, S_{h_1}(0) \geq 0, I_{h_1}(0) \geq 0, S_{h_2}(0) \geq 0, I_{h_2}(0) \geq 0$, with the total population sizes for sheep $N_s(0) = S_s(0) + E_s(0) + I_s(0) + V_s(0)$ and human beings $N_h(0) = S_{h_1}(0) + I_{h_1}(0) + S_{h_2}(0) + I_{h_2}(0)$.

2.2. Boundedness of the solution

Theorem 2.1. *The region*

$$\Omega = \left\{ (N_s(t), W(t), N_h(t)) \in \mathbb{R}_+^3 : N_s(t) \leq \frac{\Lambda_s}{\mu_s}, W(t) \leq \frac{\kappa \Lambda_s}{\mu_s(\tau + \eta\varepsilon)}, N_h(t) \leq \frac{\Lambda_h}{\min\{\mu_{h_1}, \mu_{h_2}\}} \right\}$$

is positively invariant for models (1)–(2).

Proof. Since $N_s(t) = S_s(t) + E_s(t) + I_s(t) + V_s(t)$, we obtain the equation

$$\frac{dN_s(t)}{dt} = \Lambda_s - \mu_s(S_s(t) + E_s(t) + I_s(t) + V_s(t)) - \alpha_s I_s(t) - \zeta_s I_s(t) \leq \Lambda_s - \mu_s N_s(t).$$

Thus, we get

$$N_s(t) \leq N_s(0)e^{-\mu_s t} + \frac{\Lambda_s}{\mu_s}(1 - e^{-\mu_s t}).$$

When $t \rightarrow \infty$, $N_s(t) \rightarrow \frac{\Lambda_s}{\mu_s}$, then $N_s(t)$ is bounded. Similarly, we prove that $W(t)$ and $N_h(t)$ are bounded. Therefore, the region Ω is positively invariant for models (1)–(2).

2.3. Basic reproduction number and equilibrium points

Let the right-hand side of all equations in models (1)–(2) be zero, and we obtain that the disease-free equilibrium point is $\mathbb{E}^0 = (S_s^0, 0, 0, V_s^0, 0, S_{h_1}^0, 0, S_{h_2}^0, 0)$, where

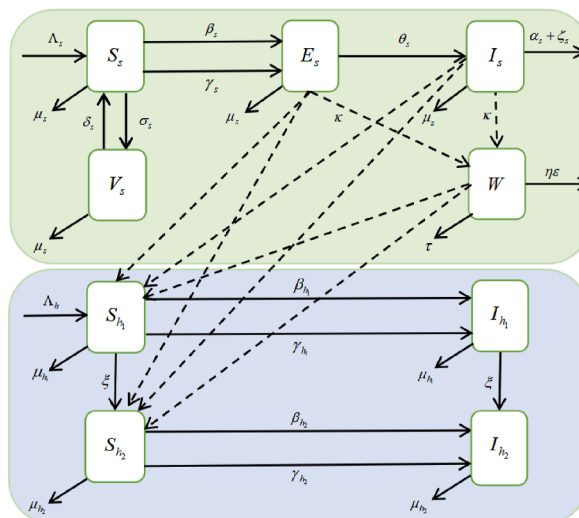


Fig. 1. The brucellosis transmission mechanism within sheep population, brucella and human population.

$$S_s^0 = \frac{\Lambda_s(\delta_s + \mu_s)}{\mu_s(\delta_s + \sigma_s + \mu_s)}, V_s^0 = \frac{\Lambda_s \sigma_s}{\mu_s(\delta_s + \sigma_s + \mu_s)}, S_{h_1}^0 = \frac{\Lambda_h}{\mu_{h_1} + \xi}, S_{h_2}^0 = \frac{\Lambda_h \xi}{\mu_{h_2}(\mu_{h_1} + \xi)}.$$

Using the next-generation matrix method (van den Driessche & Watmough, 2002) to calculate basic reproduction number, let $\mathcal{F}$ be the set of new infections of compartments $E_s, I_s, W, I_{h_1}, I_{h_2}$, and $\mathcal{G} = \mathcal{G}^- - \mathcal{G}^+$, in which $\mathcal{G}^-$ is the transfer rate of the individuals moving out from five compartments and $\mathcal{G}^+$ is the transfer rate for the individuals entering into five compartments by all other means. Then, we write down the following expressions

$$\mathcal{F} = \begin{pmatrix} \beta_s S_s(E_s + I_s) + \gamma_s S_s W \\ 0 \\ 0 \\ \beta_{h_1} S_{h_1}((1 + \mu_s)E_s + (1 + \alpha_s + \mu_s)I_s) + \gamma_{h_1} S_{h_1} W \\ \beta_{h_2} S_{h_2}((1 + \mu_s)E_s + (1 + \alpha_s + \mu_s)I_s) + \gamma_{h_2} S_{h_2} W \end{pmatrix}, \quad \mathcal{G} = \begin{pmatrix} \theta_s E_s + \mu_s E_s \\ -\theta_s E_s + \alpha_s I_s + \zeta_s I_s + \mu_s I_s \\ -\kappa(E_s + I_s) + \tau W + \eta \epsilon W \\ \mu_{h_1} I_{h_1} + \xi I_{h_1} \\ \mu_{h_2} I_{h_2} - \xi I_{h_1} \end{pmatrix},$$

the corresponding Jacobian matrices at the disease-free equilibrium point $\mathbb{E}^0$ are

$$F = \begin{pmatrix} \beta_s S_s^0 & \beta_s S_s^0 & \gamma_s S_s^0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ \beta_{h_1} S_{h_1}^0(1 + \mu_s) & \beta_{h_1} S_{h_1}^0(1 + \alpha_s + \mu_s) & \gamma_{h_1} S_{h_1}^0 & 0 & 0 \\ \beta_{h_2} S_{h_2}^0(1 + \mu_s) & \beta_{h_2} S_{h_2}^0(1 + \alpha_s + \mu_s) & \gamma_{h_2} S_{h_2}^0 & 0 & 0 \end{pmatrix},$$

$$G = \begin{pmatrix} \theta_s + \mu_s & 0 & 0 & 0 & 0 \\ -\theta_s & \alpha_s + \zeta_s + \mu_s & 0 & 0 & 0 \\ -\kappa & -\kappa & \tau + \eta \epsilon & 0 & 0 \\ 0 & 0 & 0 & \mu_{h_1} + \xi & 0 \\ 0 & 0 & 0 & -\xi & \mu_{h_2} \end{pmatrix}.$$

Thus, the basic reproduction number of models (1)–(2) is spectral radius of the matrix FG^{-1} . That is,

$$R_0 = \rho(FG^{-1}) = \frac{\beta_s S_s^0}{\theta_s + \mu_s} + \frac{\beta_s S_s^0 \theta_s}{(\theta_s + \mu_s)(\alpha_s + \zeta_s + \mu_s)} + \frac{\gamma_s S_s^0}{\tau + \eta \epsilon} \left(\frac{\kappa \theta_s}{(\theta_s + \mu_s)(\alpha_s + \zeta_s + \mu_s)} + \frac{\kappa}{\theta_s + \mu_s} \right). \quad (3)$$

Let the right-hand side of all equations in models (1)–(2) be zero, and we obtain that the endemic equilibrium point is $\mathbb{E}^* = (S_s^*, E_s^*, I_s^*, V_s^*, W^*, S_{h_1}^*, I_{h_1}^*, S_{h_2}^*, I_{h_2}^*)$, where

$$\begin{aligned}
S_s^* &= \frac{(\theta_s + \mu_s)(\alpha_s + \zeta_s + \mu_s)}{(\theta_s + \alpha_s + \zeta_s + \mu_s)} \cdot \frac{1}{\beta_s + \frac{\kappa\gamma_s}{\tau + \eta\varepsilon}}, E_s^* = \frac{\alpha_s + \zeta_s + \mu_s}{\theta_s} I_s^*, V_s^* = \frac{\sigma_s}{\delta_s + \mu_s} S_s^*, \\
W^* &= \frac{\kappa}{\tau + \eta\varepsilon} (E_s^* + I_s^*) = \frac{\kappa}{\tau + \eta\varepsilon} \left(1 + \frac{\alpha_s + \zeta_s + \mu_s}{\theta_s} \right) I_s^*, \\
I_s^* &= \frac{\theta_s \mu_s (\delta_s + \sigma_s + \mu_s)}{(\delta_s + \mu_s)(\theta_s + \alpha_s + \zeta_s + \mu_s)} \cdot \frac{1}{\beta_s + \frac{\kappa\gamma_s}{\tau + \eta\varepsilon}} (R_0 - 1), \\
S_{h_1}^* &= \frac{\Lambda_h}{\beta_{h_1}((1 + \mu_s)E_s^* + (1 + \alpha_s + \mu_s)I_s^*) + \gamma_{h_1}W^* + \mu_{h_1} + \xi}, \\
I_{h_1}^* &= \frac{\beta_{h_1}((1 + \mu_s)E_s^* + (1 + \alpha_s + \mu_s)I_s^*) + \gamma_{h_1}W^*}{\mu_{h_1} + \xi} S_{h_1}^*, \\
S_{h_2}^* &= \frac{\xi}{\beta_{h_2}((1 + \mu_s)E_s^* + (1 + \alpha_s + \mu_s)I_s^*) + \gamma_{h_2}W^* + \mu_{h_2}} S_{h_1}^*, \\
I_{h_2}^* &= \frac{\xi}{\mu_{h_2}} I_{h_1}^* + \frac{\beta_{h_2}((1 + \mu_s)E_s^* + (1 + \alpha_s + \mu_s)I_s^*) + \gamma_{h_2}W^*}{\mu_{h_2}} S_{h_2}^*.
\end{aligned}$$

If $R_0 > 1$, then $I_s^* > 0$. We further obtain $E_s^* > 0$ and $W^* > 0$. Since $S_s^* > 0$, we further derive $V_s^* > 0$. The substitution gives that $S_{h_1}^* > 0$, $I_{h_1}^* > 0$, $S_{h_2}^* > 0$, and $I_{h_2}^* > 0$. So, a unique endemic equilibrium point exists.

3. Stability analysis

Theorem 3.1. If $R_0 < 1$, then the disease-free equilibrium point $\mathbb{E}^0$ is locally asymptotically stable.

Proof. Rewriting models (1)–(2) in terms of $E_s, I_s, W, I_{h_1}, I_{h_2}, S_s, V_s, S_{h_1}, S_{h_2}$, the Jacobian matrix of the disease-free equilibrium point $\mathbb{E}^0$ is obtained as

$$J(\mathbb{E}^0) = \begin{pmatrix} J_1 & 0 \\ J_2 & J_3 \end{pmatrix},$$

where

$$J_1 = \begin{pmatrix} \beta_s S_s^0 - \theta_s - \mu_s & \beta_s S_s^0 & \gamma_s S_s^0 & 0 & 0 \\ \theta_s & -(\alpha_s + \zeta_s + \mu_s) & 0 & 0 & 0 \\ \kappa & \kappa & -(\tau + \eta\varepsilon) & 0 & 0 \\ \beta_{h_1} S_{h_1}^0 (1 + \mu_s) & \beta_{h_1} S_{h_1}^0 (1 + \alpha_s + \mu_s) & \gamma_{h_1} S_{h_1}^0 & -(\mu_{h_1} + \xi) & 0 \\ \beta_{h_2} S_{h_2}^0 (1 + \mu_s) & \beta_{h_2} S_{h_2}^0 (1 + \alpha_s + \mu_s) & \gamma_{h_2} S_{h_2}^0 & \xi & -\mu_{h_2} \end{pmatrix},$$

$$J_3 = \begin{pmatrix} -(\sigma_s + \mu_s) & \delta_s & 0 & 0 \\ \sigma_s & -(\delta_s + \mu_s) & 0 & 0 \\ 0 & 0 & -(\mu_{h_1} + \xi) & 0 \\ 0 & 0 & \xi & -\mu_{h_2} \end{pmatrix}.$$

The negativity of the eigenvalues of $J(\mathbb{E}^0)$ depends on the negativity of the eigenvalues of J_1 and J_3 . Since $|\lambda \mathbb{I}_4 - J_3| = (\lambda + \mu_{h_2})(\lambda + \mu_{h_1} + \xi)(\lambda + \delta_s + \sigma_s + \mu_s)(\lambda + \mu_s)$, where $\mathbb{I}_4$ denotes the 4×4 identity matrix, it follows that $\lambda_1 = -\mu_{h_2} < 0$, $\lambda_2 = -(\mu_{h_1} + \xi) < 0$, $\lambda_3 = -(\delta_s + \sigma_s + \mu_s) < 0$, and $\lambda_4 = -\mu_s < 0$, which implies that all eigenvalues of J_3 have negative real parts. For matrix J_1 , since $|\lambda \mathbb{I}_5 - J_1| = (\lambda + \mu_{h_2})(\lambda + \mu_{h_1} + \xi)|\lambda \mathbb{I}_3 - A|$ with

$$A = \begin{pmatrix} \beta_s S_s^0 - \theta_s - \mu_s & \beta_s S_s^0 & \gamma_s S_s^0 \\ \theta_s & -(\alpha_s + \zeta_s + \mu_s) & 0 \\ \kappa & \kappa & -(\tau + \eta\epsilon) \end{pmatrix},$$

we need to confirm all eigenvalues of A . Let $a = \beta_s S_s^0 - \theta_s - \mu_s$, $b = -(\alpha_s + \zeta_s + \mu_s)$, and $c = -(\tau + \eta\epsilon)$, and we obtain

$$|\lambda \mathbb{I}_3 - A| = \begin{vmatrix} \lambda - a & -\beta_s S_s^0 & -\gamma_s S_s^0 \\ -\theta_s & \lambda - b & 0 \\ -\kappa & -\kappa & \lambda - c \end{vmatrix} = \lambda^3 + a_1 \lambda^2 + a_2 \lambda + a_3,$$

where $a_0 = 1$, $a_1 = -(a + b + c)$, $a_2 = ac + ab + bc - \beta_s S_s^0 \theta_s - \kappa \gamma_s S_s^0$, and $a_3 = \kappa \gamma_s S_s^0 b + \beta_s S_s^0 \theta_s c - \kappa \gamma_s S_s^0 \theta_s - abc$. Since $R_0 < 1$, we get

$$(\theta_s + \mu_s) - \beta_s S_s^0 > 0, \quad (\theta_s + \mu_s - \beta_s S_s^0)(\alpha_s + \zeta_s + \mu_s) - \beta_s S_s^0 \theta_s > 0, \quad (\theta_s + \mu_s - \beta_s S_s^0)(\tau + \eta\epsilon) - \kappa \gamma_s S_s^0 > 0,$$

$$(\theta_s + \mu_s - \beta_s S_s^0)(\alpha_s + \zeta_s + \mu_s)(\tau + \eta\epsilon) - \kappa \gamma_s S_s^0(\alpha_s + \zeta_s + \mu_s) - \beta_s S_s^0 \theta_s(\tau + \eta\epsilon) - \kappa \gamma_s S_s^0 \theta_s > 0.$$

Therefore, $a_1 > 0$, $a_2 > 0$, $a_3 > 0$. Hence, $\Delta_1 = a_1 > 0$, $\Delta_2 = a_1 a_2 - a_3 > 0$, $\Delta_3 = a_3 \Delta_2 > 0$. By Routh-Hurwitz Criterion, all eigenvalues of A have negative real parts. Since J_1 additionally has two negative eigenvalues $-\mu_{h_2}$ and $-(\mu_{h_1} + \xi)$, all eigenvalues of J_1 have negative real parts. Consequently, all eigenvalues of $J(\mathbb{E}^0)$ have negative real parts. Therefore, if $R_0 < 1$, then the disease-free equilibrium point $\mathbb{E}^0$ is locally asymptotically stable.

Theorem 3.2. If $R_0 < 1$, then the disease-free equilibrium point $\mathbb{E}^0$ is globally asymptotically stable.

Proof. Step 1. The disease-free equilibrium point of model (1) is $\mathbb{E}_s^0 = (S_s^0, 0, 0, V_s^0, 0)$. Define a Lyapunov function

$$L_1 = \left(S_s - S_s^0 - S_s^0 \ln \frac{S_s}{S_s^0} \right) + \left(V_s - V_s^0 - V_s^0 \ln \frac{V_s}{V_s^0} \right) + E_s + a_1 I_s + a_2 W,$$

where

$$a_1 = \frac{1}{\alpha_s + \zeta_s + \mu_s} \left(\beta_s S_s^0 + \frac{\kappa \gamma_s S_s^0}{\tau + \eta\epsilon} \right), \quad a_2 = \frac{\gamma_s S_s^0}{\tau + \eta\epsilon}.$$

Taking the derivative of L_1 , by using the equalities for $\mathbb{E}^0$, we obtain

$$\begin{aligned}
\frac{dL_1}{dt} &= \left(1 - \frac{S_s^0}{S_s}\right) [\Lambda_s - \beta_s S_s (E_s + I_s) - \gamma_s S_s W + \delta_s V_s - \sigma_s S_s - \mu_s S_s] \\
&\quad + \left(1 - \frac{V_s^0}{V_s}\right) (\sigma_s S_s - \delta_s V_s - \mu_s V_s) + [\beta_s S_s (E_s + I_s) + \gamma_s S_s W - \theta_s E_s - \mu_s E_s] \\
&\quad + a_1 (\theta_s E_s - \alpha_s I_s - \zeta_s I_s - \mu_s I_s) + a_2 [\kappa (E_s + I_s) - (\tau + \eta \epsilon) W] \\
&= \left(1 - \frac{S_s^0}{S_s}\right) \left(\Lambda_s + \delta_s V_s - \frac{\Lambda_s + \delta_s V_s^0}{S_s^0} S_s\right) + \left(1 - \frac{V_s^0}{V_s}\right) \left(\sigma_s S_s - \frac{\sigma_s S_s^0}{V_s^0} V_s\right) \\
&\quad - \left(1 - \frac{S_s^0}{S_s}\right) \beta_s S_s E_s + \beta_s S_s E_s - \theta_s E_s - \mu_s E_s + a_1 \theta_s E_s + a_2 \kappa E_s \\
&= \Lambda_s \left(2 - \frac{S_s}{S_s^0} - \frac{S_s^0}{S_s}\right) + \delta_s (S_s - S_s^0) \left(\frac{V_s}{S_s} - \frac{V_s^0}{S_s^0}\right) + \sigma_s (V_s - V_s^0) \left(\frac{S_s}{V_s} - \frac{S_s^0}{V_s^0}\right) \\
&\quad + \frac{S_s^0}{S_s} \beta_s S_s E_s - (\theta_s + \mu_s) E_s + a_1 \theta_s E_s + a_2 \kappa E_s \\
&= (\mu_s S_s^0 + \mu_s V_s^0) \left(2 - \frac{S_s}{S_s^0} - \frac{S_s^0}{S_s}\right) + \delta_s (S_s - S_s^0) \left(\frac{V_s}{S_s} - \frac{V_s^0}{S_s^0}\right) + \frac{\delta_s + \mu_s V_s^0}{S_s^0} (V_s - V_s^0) \left(\frac{S_s}{V_s} - \frac{S_s^0}{V_s^0}\right) \\
&\quad + (\theta_s + \mu_s) \left(\frac{\beta_s S_s^0}{\theta_s + \mu_s} - 1 + \frac{\theta_s}{(\theta_s + \mu_s)(\alpha_s + \zeta_s + \mu_s)} \left(\beta_s S_s^0 + \frac{\kappa \gamma_s S_s^0}{\tau + \eta \epsilon}\right) + \frac{\kappa \gamma_s S_s^0}{(\theta_s + \mu_s)(\tau + \eta \epsilon)}\right) E_s \\
&= \mu_s S_s^0 \left(2 - \frac{S_s}{S_s^0} - \frac{S_s^0}{S_s}\right) + \mu_s V_s^0 \left(3 - \frac{S_s^0}{S_s} - \frac{V_s}{V_s^0} - \frac{S_s V_s^0}{S_s^0 V_s}\right) + \delta_s V_s^0 \left(2 - \frac{S_s^0 V_s}{S_s V_s^0} - \frac{S_s V_s^0}{S_s^0 V_s}\right) \\
&\quad + (\theta_s + \mu_s) (R_0 - 1) E_s.
\end{aligned}$$

Since $x_1 + x_2 + \dots + x_n \geq n \sqrt[n]{x_1 x_2 \dots x_n}$ for $x_1 \geq 0, x_2 \geq 0, \dots, x_n \geq 0$, we obtain that

$$2 - \frac{S_s}{S_s^0} - \frac{S_s^0}{S_s} \leq 0, \quad 3 - \frac{S_s^0}{S_s} - \frac{V_s}{V_s^0} - \frac{S_s V_s^0}{S_s^0 V_s} \leq 0, \quad 2 - \frac{S_s^0 V_s}{S_s V_s^0} - \frac{S_s V_s^0}{S_s^0 V_s} \leq 0.$$

If $R_0 < 1$, it then follows that $\frac{dL_1}{dt} \leq 0$. For $\frac{dL_1}{dt} = 0$, we derive that $S_s = S_s^0$, $V_s = V_s^0$, and $E_s = 0$. Let $\Omega_1 = \{(S_s, E_s, I_s, V_s, W) \in \mathbb{R}_+^5 | \frac{dL_1}{dt} = 0\} = \{\mathbb{E}_s^0 = (S_s^0, 0, 0, V_s^0, 0)\}$. By substituting $S_s = S_s^0$, $V_s = V_s^0$, and $E_s = 0$ into model (1), we obtain that $I_s = 0$ and $W = 0$. Thus, $\{\mathbb{E}_s^0\}$ is the largest invariant set in Ω_1 . According to LaSalle's Invariant Principle, $\mathbb{E}_s^0$ is globally asymptotically stable.

Step 2. We substitute the disease-free equilibrium point $\mathbb{E}_s^0$ into model (2), and derive

$$\begin{cases} \frac{dS_{h_1}}{dt} = \Lambda_h - \mu_{h_1} S_{h_1} - \xi S_{h_1}, \\ \frac{dI_{h_1}}{dt} = -\mu_{h_1} I_{h_1} - \xi I_{h_1}, \\ \frac{dS_{h_2}}{dt} = \xi S_{h_1} - \mu_{h_2} S_{h_2}, \\ \frac{dI_{h_2}}{dt} = \xi I_{h_1} - \mu_{h_2} I_{h_2}. \end{cases} \quad (4)$$

The disease-free equilibrium point of model (4) is $\mathbb{E}_h^0 = (S_{h_1}^0, 0, S_{h_2}^0, 0)$. Define a Lyapunov function

$$L_2 = \left(S_{h_1} - S_{h_1}^0 - S_{h_1}^0 \ln \frac{S_{h_1}}{S_{h_1}^0} \right) + I_{h_1} + I_{h_2}.$$

Taking the derivative of L_2 , we obtain

$$\frac{dL_2}{dt} = -\frac{\mu_{h_1} + \xi}{S_{h_1}} (S_{h_1} - S_{h_1}^0)^2 - \mu_{h_1} I_{h_1} - \mu_{h_2} I_{h_2}.$$

Therefore, it then follows that $\frac{dL_2}{dt} \leq 0$. For $\frac{dL_2}{dt} = 0$, we obtain $S_{h_1} = S_{h_1}^0$, $I_{h_1} = 0$, and $I_{h_2} = 0$. Let $\Omega_2 = \left\{ (S_{h_1}, I_{h_1}, S_{h_2}, I_{h_2}) \in \mathbb{R}_+^4 \mid \frac{dL_2}{dt} = 0 \right\} = \left\{ \mathbb{E}_h^0 = (S_{h_1}^0, 0, S_{h_2}^0, 0) \right\}$. By substituting $S_{h_1} = S_{h_1}^0$, $I_{h_1} = 0$, and $I_{h_2} = 0$ into model (4), we obtain that $S_{h_2} = S_{h_2}^0$. Thus, $\{\mathbb{E}_h^0\}$ is the largest invariant set in Ω_2 . According to LaSalle's Invariance Principle, the disease-free equilibrium point $\mathbb{E}_h^0$ of model (4) is globally asymptotically stable. By asymptotically autonomous system theory, we obtain that the disease-free equilibrium point $\mathbb{E}_h^0$ of model (2) is globally asymptotically stable. Consequently, both $\mathbb{E}_s^0$ and $\mathbb{E}_h^0$ are globally asymptotically stable, and it follows that the disease-free equilibrium point $\mathbb{E}^0$ of models (1)–(2) is globally asymptotically stable.

Theorem 3.3. If $R_0 > 1$, then the endemic equilibrium point $\mathbb{E}^*$ is globally asymptotically stable.

Proof. Step 1. The endemic equilibrium point of model (1) is $\mathbb{E}_{ss}^* = (S_s^*, E_s^*, I_s^*, V_s^*, W^*)$, and by using the equalities for $\mathbb{E}_{ss}^*$, model (1) is transformed as

$$\begin{cases} \frac{dS_s}{dt} = \frac{\Lambda_s}{S_s} S_s \left(\frac{S_s^*}{S_s} - 1 \right) - \beta_s S_s E_s^* \left(\frac{E_s^*}{E_s^*} - 1 \right) - \beta_s S_s I_s^* \left(\frac{I_s^*}{I_s^*} - 1 \right) - \gamma_s S_s W^* \left(\frac{W}{W^*} - 1 \right) + \frac{\delta_s V_s^*}{S_s} S_s \left(\frac{S_s^* V_s^*}{S_s V_s^*} - 1 \right), \\ \frac{dE_s}{dt} = \frac{\beta_s S_s^* E_s^*}{E_s^*} E_s \left(\frac{S_s E_s^*}{S_s^* E_s^*} - 1 \right) + \frac{\beta_s S_s^* I_s^*}{E_s^*} E_s \left(\frac{S_s E_s^* I_s^*}{S_s^* E_s^* I_s^*} - 1 \right) + \frac{\gamma_s S_s^* W^*}{E_s^*} E_s \left(\frac{S_s E_s^* W}{S_s^* E_s^* W^*} - 1 \right), \\ \frac{dI_s}{dt} = \frac{\theta_s E_s^*}{I_s^*} I_s \left(\frac{E_s I_s^*}{E_s^* I_s^*} - 1 \right), \\ \frac{dV_s}{dt} = \frac{\sigma_s S_s^*}{V_s^*} V_s \left(\frac{S_s V_s^*}{S_s^* V_s^*} - 1 \right), \\ \frac{dW}{dt} = \frac{\kappa E_s^*}{W^*} W \left(\frac{E_s W^*}{E_s^* W^*} - 1 \right) + \frac{\kappa I_s^*}{W^*} W \left(\frac{I_s W^*}{I_s^* W^*} - 1 \right). \end{cases}$$

Define a Lyapunov function

$$\begin{aligned} L_3 = & \left(S_s - S_s^* - S_s^* \ln \frac{S_s}{S_s^*} \right) + \left(V_s - V_s^* - V_s^* \ln \frac{V_s}{V_s^*} \right) + \left(E_s - E_s^* - E_s^* \ln \frac{E_s}{E_s^*} \right) \\ & + a_1 \left(I_s - I_s^* - I_s^* \ln \frac{I_s}{I_s^*} \right) + a_2 \left(W - W^* - W^* \ln \frac{W}{W^*} \right), \end{aligned}$$

where

$$a_1 = \frac{\beta_s S_s^* I_s^*}{\theta_s E_s^*} + \frac{\gamma_s S_s^* I_s^* W^*}{\theta_s E_s^* (E_s^* + I_s^*)}, \quad a_2 = \frac{\gamma_s S_s^* W^*}{\kappa (E_s^* + I_s^*)}.$$

Taking the derivative of L_3 , we derive

$$\begin{aligned}
\frac{dL_3}{dt} &= \left(1 - \frac{S_s^*}{S_s}\right) \left(\frac{\Lambda_s}{S_s^*} S_s \left(\frac{S_s^*}{S_s} - 1\right) - \beta_s S_s E_s^* \left(\frac{E_s}{E_s^*} - 1\right) - \beta_s S_s I_s^* \left(\frac{I_s}{I_s^*} - 1\right) - \gamma_s S_s W^* \left(\frac{W}{W^*} - 1\right) \right. \\
&\quad \left. + \frac{\delta_s V_s^*}{S_s^*} S_s \left(\frac{S_s^* V_s}{S_s V_s^*} - 1\right) \right) + \left(1 - \frac{V_s^*}{V_s}\right) \left(\frac{\sigma_s S_s^*}{V_s^*} V_s \left(\frac{S_s V_s^*}{S_s^* V_s} - 1\right) \right) \\
&\quad + \left(1 - \frac{E_s^*}{E_s}\right) \left(\beta_s S_s^* E_s \left(\frac{S_s}{S_s^*} - 1\right) + \frac{\beta_s S_s^* I_s^*}{E_s^*} E_s \left(\frac{S_s I_s}{S_s^* I_s^*} - 1\right) + \frac{\gamma_s S_s^* W^*}{E_s^*} E_s \left(\frac{S_s W}{S_s^* W^*} - 1\right) \right) \\
&\quad + a_1 \left(1 - \frac{I_s^*}{I_s}\right) \left(\frac{\theta_s E_s^*}{I_s^*} I_s \left(\frac{E_s I_s^*}{E_s^* I_s} - 1\right) \right) + a_2 \left(1 - \frac{W^*}{W}\right) \left(\frac{\kappa E_s^*}{W^*} W \left(\frac{E_s W^*}{E_s^* W} - 1\right) + \frac{\kappa I_s^*}{W^*} W \left(\frac{I_s W^*}{I_s^* W} - 1\right) \right) \\
&= [\beta_s S_s^* (E_s^* + I_s^*) + \gamma_s S_s^* W^* + \mu_s V_s^* + \mu_s S_s^*] \left(2 - \frac{S_s}{S_s^*} - \frac{S_s^*}{S_s}\right) + \delta_s (S_s - S_s^*) \left(\frac{V_s}{S_s} - \frac{V_s^*}{S_s^*}\right) \\
&\quad + \frac{\delta_s + \mu_s V_s^*}{S_s^*} (V_s - V_s^*) \left(\frac{S_s}{V_s} - \frac{S_s^*}{V_s^*}\right) + \beta_s S_s^* I_s^* \left(\frac{S_s}{S_s^*} + \frac{I_s}{I_s^*} - \frac{E_s}{E_s^*} - \frac{S_s E_s^* I_s}{S_s^* E_s I_s^*}\right) \\
&\quad + \gamma_s S_s^* W^* \left(\frac{S_s}{S_s^*} + \frac{W}{W^*} - \frac{E_s}{E_s^*} - \frac{S_s E_s^* W}{S_s^* E_s W^*}\right) + a_1 \theta_s E_s^* \left(\frac{E_s}{E_s^*} - \frac{I_s}{I_s^*} - \frac{E_s I_s^*}{E_s^* I_s} + 1\right) \\
&\quad + \kappa a_2 E_s^* \left(\frac{E_s}{E_s^*} - \frac{W}{W^*} - \frac{E_s W^*}{E_s^* W} + 1\right) + \kappa a_2 I_s^* \left(\frac{I_s}{I_s^*} - \frac{W}{W^*} - \frac{I_s W^*}{I_s^* W} + 1\right) \\
&= (\beta_s S_s^* E_s^* + \mu_s S_s^*) \left(2 - \frac{S_s}{S_s^*} - \frac{S_s^*}{S_s}\right) + \mu_s V_s^* \left(3 - \frac{S_s}{S_s^*} - \frac{V_s}{V_s^*} - \frac{S_s V_s^*}{S_s^* V_s}\right) + \delta_s V_s^* \left(2 - \frac{S_s V_s^*}{S_s^* V_s} - \frac{S_s V_s^*}{S_s^* V_s}\right) \\
&\quad + \beta_s S_s^* I_s^* \left(3 - \frac{S_s}{S_s^*} - \frac{E_s I_s^*}{E_s^* I_s} - \frac{S_s E_s^* I_s}{S_s^* E_s I_s^*}\right) + \frac{\gamma_s S_s^* E_s^* W^*}{E_s^* + I_s^*} \left(3 - \frac{S_s^*}{S_s} - \frac{E_s W^*}{E_s^* W} - \frac{S_s E_s^* W}{S_s^* E_s W^*}\right) \\
&\quad + \frac{\gamma_s S_s^* I_s^* W^*}{E_s^* + I_s^*} \left(4 - \frac{S_s^*}{S_s} - \frac{E_s I_s^*}{E_s^* I_s} - \frac{I_s W^*}{I_s^* W} - \frac{S_s E_s^* W}{S_s^* E_s W^*}\right).
\end{aligned}$$

Since $x_1 + x_2 + \dots + x_n \geq n\sqrt[n]{x_1 x_2 \dots x_n}$ for $x_1 \geq 0, x_2 \geq 0, \dots, x_n \geq 0$, we obtain that

$$\begin{aligned}
2 - \frac{S_s}{S_s^*} - \frac{S_s^*}{S_s} &\leq 0, \quad 3 - \frac{S_s^*}{S_s} - \frac{V_s}{V_s^*} - \frac{S_s V_s^*}{S_s^* V_s} \leq 0, \\
2 - \frac{S_s^* V_s}{S_s V_s^*} - \frac{S_s V_s^*}{S_s^* V_s} &\leq 0, \quad 3 - \frac{S_s^*}{S_s} - \frac{E_s I_s^*}{E_s^* I_s} - \frac{S_s E_s^* I_s}{S_s^* E_s I_s^*} \leq 0, \\
3 - \frac{S_s^*}{S_s} - \frac{E_s W^*}{E_s^* W} - \frac{S_s E_s^* W}{S_s^* E_s W^*} &\leq 0, \quad 4 - \frac{S_s^*}{S_s} - \frac{E_s I_s^*}{E_s^* I_s} - \frac{I_s W^*}{I_s^* W} - \frac{S_s E_s^* W}{S_s^* E_s W^*} \leq 0.
\end{aligned}$$

If $R_0 > 1$, it then follows that $\frac{dL_3}{dt} \leq 0$. For $\frac{dL_3}{dt} = 0$, we obtain that $S_s = S_s^*$ and $V_s = V_s^*$. Let $\Omega_3 = \left\{ (S_s, E_s, I_s, V_s, W) \in \mathbb{R}_+^5 \mid \frac{dL_3}{dt} = 0 \right\} = \{ \mathbb{E}_{ss}^* = (S_s^*, E_s^*, I_s^*, V_s^*, W^*) \}$. By substituting $S_s = S_s^*$ and $V_s = V_s^*$ into model (1), we obtain that $E_s = E_s^*, I_s = I_s^*$ and $W = W^*$. Thus, $\mathbb{E}_{ss}^*$ is the largest invariant set in Ω_3 . According to LaSalle's Invariant Principle, $\mathbb{E}_{ss}^*$ is globally asymptotically stable.

Step 2. We substitute the endemic equilibrium point $\mathbb{E}_{ss}^*$ into model (2), and derive

$$\begin{cases} \frac{dS_{h_1}}{dt} = \Lambda_h - \beta_{h_1} S_{h_1} ((1 + \mu_s) E_s^* + (1 + \alpha_s + \mu_s) I_s^*) - \gamma_{h_1} S_{h_1} W^* - \mu_{h_1} S_{h_1} - \xi S_{h_1}, \\ \frac{dI_{h_1}}{dt} = \beta_{h_1} S_{h_1} ((1 + \mu_s) E_s^* + (1 + \alpha_s + \mu_s) I_s^*) + \gamma_{h_1} S_{h_1} W^* - \mu_{h_1} I_{h_1} - \xi I_{h_1}, \\ \frac{dS_{h_2}}{dt} = \xi S_{h_1} - \beta_{h_2} S_{h_2} ((1 + \mu_s) E_s^* + (1 + \alpha_s + \mu_s) I_s^*) - \gamma_{h_2} S_{h_2} W^* - \mu_{h_2} S_{h_2}, \\ \frac{dI_{h_2}}{dt} = \xi I_{h_1} + \beta_{h_2} S_{h_2} ((1 + \mu_s) E_s^* + (1 + \alpha_s + \mu_s) I_s^*) + \gamma_{h_2} S_{h_2} W^* - \mu_{h_2} I_{h_2}, \end{cases} \quad (5)$$

which gives the endemic equilibrium point $\mathbb{E}_h^* = (S_{h_1}^*, I_{h_1}^*, S_{h_2}^*, I_{h_2}^*)$ of model (5). Define a Lyapunov function

$$L_4 = \left(S_{h_1} - S_{h_1}^* - S_{h_1}^* \ln \frac{S_{h_1}}{S_{h_1}^*} \right) + \left(I_{h_1} - I_{h_1}^* - I_{h_1}^* \ln \frac{I_{h_1}}{I_{h_1}^*} \right).$$

Taking the derivative of L_4 , we derive

$$\begin{aligned} \frac{dL_4}{dt} &= \left(1 - \frac{S_{h_1}^*}{S_{h_1}} \right) [\Lambda_h - \beta_{h_1} S_{h_1} ((1 + \mu_s) E_s^* + (1 + \alpha_s + \mu_s) I_s^*) - \gamma_{h_1} S_{h_1} W^* - \mu_{h_1} S_{h_1} - \xi S_{h_1}] \\ &\quad + \left(1 - \frac{I_{h_1}^*}{I_{h_1}} \right) [\beta_{h_1} S_{h_1} ((1 + \mu_s) E_s^* + (1 + \alpha_s + \mu_s) I_s^*) + \gamma_{h_1} S_{h_1} W^* - \mu_{h_1} I_{h_1} - \xi I_{h_1}] \\ &= [\beta_{h_1} S_{h_1}^* ((1 + \mu_s) E_s^* + (1 + \alpha_s + \mu_s) I_s^*) + \gamma_{h_1} S_{h_1}^* W^* + \mu_{h_1} S_{h_1}^* + \xi S_{h_1}^*] \left(2 - \frac{S_{h_1}^*}{S_{h_1}} - \frac{S_{h_1}}{S_{h_1}^*} \right) \\ &\quad + [\beta_{h_1} S_{h_1}^* ((1 + \mu_s) E_s^* + (1 + \alpha_s + \mu_s) I_s^*) + \gamma_{h_1} S_{h_1}^* W^*] \left(\frac{S_{h_1}}{S_{h_1}^*} - \frac{I_{h_1}}{I_{h_1}^*} - \frac{S_{h_1} I_{h_1}^*}{S_{h_1}^* I_{h_1}} + 1 \right) \\ &= [\beta_{h_1} S_{h_1}^* ((1 + \mu_s) E_s^* + (1 + \alpha_s + \mu_s) I_s^*) + \gamma_{h_1} S_{h_1}^* W^*] \left(3 - \frac{S_{h_1}^*}{S_{h_1}} - \frac{I_{h_1}}{I_{h_1}^*} - \frac{S_{h_1} I_{h_1}^*}{S_{h_1}^* I_{h_1}} \right) \\ &\quad + (\mu_{h_1} S_{h_1}^* + \xi I_{h_1}^*) \left(2 - \frac{S_{h_1}^*}{S_{h_1}} - \frac{S_{h_1}}{S_{h_1}^*} \right) \end{aligned}$$

Since $x_1 + x_2 + \dots + x_n \geq n \sqrt[n]{x_1 x_2 \dots x_n}$ for $x_1 \geq 0, x_2 \geq 0, \dots, x_n \geq 0$, we obtain that

$$3 - \frac{S_{h_1}^*}{S_{h_1}} - \frac{I_{h_1}}{I_{h_1}^*} - \frac{S_{h_1} I_{h_1}^*}{S_{h_1}^* I_{h_1}} \leq 0, \quad 2 - \frac{S_{h_1}^*}{S_{h_1}} - \frac{S_{h_1}}{S_{h_1}^*} \leq 0.$$

Therefore, it then follows that $\frac{dL_4}{dt} \leq 0$. For $\frac{dL_4}{dt} = 0$, we obtain that $S_{h_1} = S_{h_1}^*$ and $I_{h_1} = I_{h_1}^*$. Let $\Omega_4 = \left\{ (S_{h_1}, I_{h_1}, S_{h_2}, I_{h_2}) \in \mathbb{R}_+^4 \mid \frac{dL_4}{dt} = 0 \right\} = \left\{ \mathbb{E}_h^* = (S_{h_1}^*, I_{h_1}^*, S_{h_2}^*, I_{h_2}^*) \right\}$. By substituting $S_{h_1} = S_{h_1}^*$ and $I_{h_1} = I_{h_1}^*$ into model (5), we obtain that $S_{h_2} = S_{h_2}^*$ and $I_{h_2} = I_{h_2}^*$. Thus, $\{\mathbb{E}_h^*\}$ is the largest invariant set in Ω_4 . According to LaSalle's Invariance Principle, the endemic equilibrium point $\mathbb{E}_h^*$ of model (5) is globally asymptotically stable. By asymptotically autonomous system theory, we obtain that the endemic equilibrium point $\mathbb{E}_h^*$ of model (2) is globally asymptotically stable. Consequently, both $\mathbb{E}_{ss}^*$ and $\mathbb{E}_h^*$ are globally asymptotically stable, and it follows that the endemic equilibrium point $\mathbb{E}^*$ of models (1)-(2) is globally asymptotically stable.

4. Numerical simulations

4.1. Data fitting with parameter estimation

The Chinese government implemented two control measures for preventing the spread of brucellosis in 2018 (National Health Commission of the People's Republic of China, 2018) and 2022 (Ministry of Agriculture and Rural Affairs of the People's Republic of China, 2022), and also implemented two measures for promoting the development of livestock in January 2020 (Ministry of Agriculture and Rural Affairs of the People's Republic of China, 2020) and December 2020 (Ministry of Agriculture and Rural Affairs of the People's Republic of China, 2020). The implementation of two livestock-related measures contributed to a further increase in the annual sheep breeding volume in 2020 and 2021. The period that surveillance data ranging from 2017 to 2023 is separated into five phases by four measures: January 2017 to March 2018 (P_1), April 2018 to December 2019 (P_2), January 2020 to December 2020 (P_3), January 2021 to March 2022 (P_4), and April 2022 to December 2023 (P_5). The total population of Jinzhou is divided into Group G1 (0-34 years old) and Group G2 (35 years old and above), and the population size is collected from the Statistical Bulletin of National Economic and Social Development of Jinzhou in 2016 (Jinzhou Bureau of Statistics, 2016). The 2017-2023 monthly brucellosis surveillance data with two age groups are obtained from Jinzhou Center for Disease Control and Prevention (i.e., Jinzhou CDC), of which the monthly number of brucellosis cases from 2020 to 2022 increases due to the implementation of two livestock-related measures by the Chinese government in 2020 (Fig. 2).

To describe the free and hidden transmission of brucellosis in the local population, we adopt the awareness delay, the period between the date that the individuals were infected for the first time and the date that the same individuals were

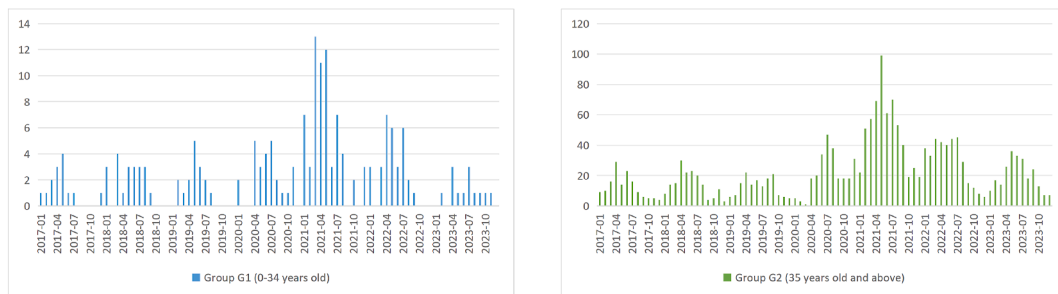


Fig. 2. Monthly surveillance data of Jinzhou brucellosis for Group G1 and Group G2 from 2017 to 2023.

firstly confirmed, to describe the insufficient surveillance for the local governments. The relevant results discover that the awareness delay is independent of epidemic models and surveillance data, and that the longer awareness delay produces the larger infection scales (Chen et al., 2024; Wang et al., 2025). Since the average incubation period of human brucellosis ranges from one week to three weeks, the awareness delay of human brucellosis is set as one-month in this study. The step size is set as 0.01 for the numerical simulations. Next, we start the parameter estimations of models (1)–(2) to determine basic reproduction numbers, infection scales, and brucellosis transmission risks. The optimal fitting of cumulative infection scales of brucellosis for two age groups is performed in Fig. 3, with the main parameter estimations from Table 1 and one-month awareness delay, by the least squares method (Chen et al., 2025; Lan et al., 2024; Wang et al., 2024; Wei et al., 2024). The values of basic reproduction numbers of brucellosis decrease in the second and fifth phases due to the implementation of two control measures in 2018 and 2022, respectively, and increase in the third and fourth phases due to the implementation of two livestock-related measures in 2020. The minimum and maximum of basic reproduction numbers respectively are reached at 1.69 in 2019 and 2.40 in 2021 as presented in Fig. 3.

The yearly cumulative infection scales of Jinzhou brucellosis epidemics with two age groups are fitted using EpiSIX (Fig. 4) and the parameter values from Table 2. The fitting results are reliable, because the RMSE values are successively below or around 5%. The consistency between the fitting results and the 2017–2023 surveillance data demonstrates the excellent fitting performance by EpiSIX. The fitting results by EpiSIX also provide the weeks that Jinzhou brucellosis epidemics are controlled. The values of basic reproduction numbers derived from the least squares method and EpiSIX (Figs. 3 and 4) are compared with the ones in (Gong et al., 2023; Liu et al., 2025; Pei et al., 2025). These results show that the values of basic reproduction numbers of our study are close to those as presented in Fig. 5.

4.2. Sensitivity analysis of main parameters

The local sensitivity analysis assesses the impacts of each parameter on basic reproduction number in Fig. 6, of which the direct transmission rate (i.e., β_s) and the constant recruitment rate (i.e., Λ_s) have significant impacts among five positive-correlation sensitive parameters, and the culling rate of infectious sheep (i.e., ζ_s) and the vaccination rate (i.e., σ_s) have significant impacts among eight negative-correlation sensitive parameters. The impacts of main parameters on the peak of infected individuals with two age groups are investigated using the Latin Hypercube Sampling (LHS) and Partial Rank Correlation Coefficients (PRCCs). The direct transmission rate (i.e., β_s) of sheep population has a greater impact than other sheep-related parameters, while the direct transmission rate (i.e., β_h) of human population has a stronger impact than other

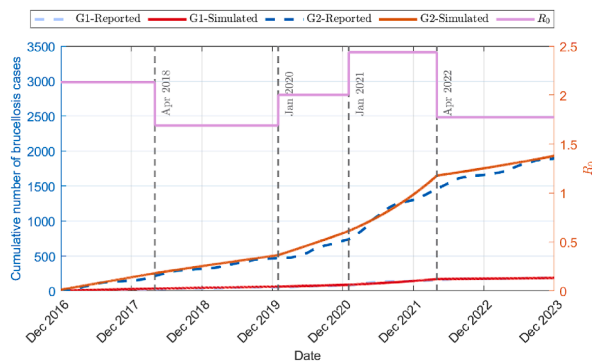


Fig. 3. Optimal fitting curves of Group G1 and Group G2 by the least squares method with one-month awareness delay, and basic reproduction numbers in phases of Jinzhou brucellosis epidemics.

Table 1
Values of main parameters of Jinzhou brucellosis epidemics from 2017 to 2023.

Parameter	Value	Period (Start - End) ^a	Source	Parameter	Value	Period (Start - End) ^a	Source
Λ_s	31,783	2016-12 – 2023-12	Estimated	Λ_h	760	2016-12 – 2023-12	(Chen et al., 2025)
β_s	1.1200×10^{-7}	2016-12 – 2018-03	Fitted	β_{h1}	8.5000×10^{-9}	2016-12 – 2018-03	Fitted
β_s	9.2200×10^{-8}	2018-04 – 2019-12	Fitted	β_{h1}	6.5000×10^{-9}	2018-04 – 2019-12	Fitted
β_s	9.5000×10^{-8}	2020-01 – 2020-12	Fitted	β_{h1}	1.0500×10^{-8}	2020-01 – 2020-12	Fitted
β_s	1.2800×10^{-7}	2021-01 – 2022-03	Fitted	β_{h1}	1.6500×10^{-8}	2021-01 – 2022-03	Fitted
β_s	7.5000×10^{-8}	2022-04 – 2023-12	Fitted	β_{h1}	1.5000×10^{-9}	2022-04 – 2023-12	Fitted
γ_s	2.6000×10^{-9}	2016-12 – 2018-03	Fitted	γ_{h1}	9.0000×10^{-11}	2016-12 – 2018-03	Fitted
γ_s	2.0000×10^{-9}	2018-04 – 2019-12	Fitted	γ_{h1}	8.0000×10^{-11}	2018-04 – 2019-12	Fitted
γ_s	2.5000×10^{-9}	2020-01 – 2020-12	Fitted	γ_{h1}	1.2000×10^{-10}	2020-01 – 2020-12	Fitted
γ_s	3.2000×10^{-9}	2021-01 – 2022-03	Fitted	γ_{h1}	1.4000×10^{-10}	2021-01 – 2022-03	Fitted
γ_s	2.4000×10^{-9}	2022-04 – 2023-12	Fitted	γ_{h1}	6.0000×10^{-11}	2022-04 – 2023-12	Fitted
δ_s	0.0330	2016-12 – 2023-12	(Hou et al., 2013)	β_{h2}	2.9000×10^{-8}	2016-12 – 2018-03	Fitted
σ_s	0.0600	2016-12 – 2018-03	Fitted	β_{h2}	2.5400×10^{-8}	2018-04 – 2019-12	Fitted
σ_s	0.0640	2018-04 – 2019-12	Fitted	β_{h2}	5.6600×10^{-8}	2020-01 – 2020-12	Fitted
σ_s	0.0500	2020-01 – 2020-12	Fitted	β_{h2}	6.5600×10^{-8}	2021-01 – 2022-03	Fitted
σ_s	0.0600	2021-01 – 2022-03	Fitted	β_{h2}	1.1000×10^{-8}	2022-04 – 2023-12	Fitted
σ_s	0.0400	2022-04 – 2023-12	Fitted	γ_{h2}	1.8000×10^{-10}	2016-12 – 2018-03	Fitted
μ_s	0.0109	2016-12 – 2023-12	Data ^b	γ_{h2}	1.6000×10^{-10}	2018-04 – 2019-12	Fitted
α_s	0.0090	2016-12 – 2023-12	Fitted ^e	γ_{h2}	2.4000×10^{-10}	2020-01 – 2020-12	Fitted
ζ_s	0.0490	2016-12 – 2023-12	Fitted ^e	γ_{h2}	2.8000×10^{-10}	2021-01 – 2022-03	Fitted
κ	1.2500	2016-12 – 2023-12	(Hou et al., 2013)	γ_{h2}	1.2000×10^{-10}	2022-04 – 2023-12	Fitted
τ	0.3000	2016-12 – 2023-12	(Hou et al., 2013)	μ_{h1}	2.5900×10^{-5}	2016-12 – 2023-12	Data ^b
ε	0.8000	2016-12 – 2023-12	Fitted ^e	ξ	2.4510×10^{-3}	2016-12 – 2023-12	Estimated
θ_s	1	2016-12 – 2023-12	Fitted	μ_{h2}	1.1500×10^{-3}	2016-12 – 2023-12	Data ^b
η	5	2016-12 – 2023-12	Fitted ^e	–	–	–	–

^a 2016 - 12 is denoted in terms of YYYY - MM, where Y is for year and M is for month.

^b Data is from Jinzhou CDC.

^c The total population size of sheep is $N_s(0) = 473,015$ from Jinzhou CDC, and the total population size of human beings is $N_h(0) = 3,022,000$ from Jinzhou Bureau of Statistics (Jinzhou Bureau of Statistics, 2018). Based on basic reproduction numbers 1.86 for years 2010-2020 (Pei et al., 2025), and 2.83 for years 2001-2020 (Liu et al., 2025), we assume that basic reproduction number 2.00 is for 2017 estimation in this study, combining with the surveillance data in 2017, we then derive that the initial scales for Group G1 and Group G2 are respectively $I_{h1}(0) = 1 \times 2.00 = 2$ and $I_{h2}(0) = 9 \times 2.00 = 18$. By the data from Jinzhou Bureau of Statistics (Jinzhou Bureau of Statistics, 2018), the average proportions of Group G1 and Group G2 are 0.31 and 0.69 respectively, thus, the corresponding initial scales are $S_{h1}(0) = (N_h(0) - I_{h1}(0) - I_{h2}(0)) \times 0.31 = 936,814$ and $S_{h2}(0) = (N_h(0) - I_{h1}(0) - I_{h2}(0)) \times 0.69 = 2,085,166$. By the ratio 12:1 for infected sheep cases to human cases (Liu et al., 2024), we derive the initial scale of sheep population $E_s(0) + I_s(0) = (I_{h1}(0) + I_{h2}(0)) \times 12 = 240$ with $E_s(0) = 180$, $I_s(0) = 60$. The initial scale of vaccinated sheep population is given by $(\text{Immunized sheep population size}) \times (\text{Annual stock size}) / (\text{Annual feeding size})$, that is $V_s(0) = 380,998$, and susceptible sheep population $S_s(0) = N_s(0) - E_s(0) - I_s(0) - V_s(0) = 91,777$.

^d We assume that the initial scale of the number of brucella in environment is $W(0) = 10,000$ in this study.

^e The 95% confidence intervals for the five parameters α_s , ζ_s , ε , θ_s and η are [0.0068,0.0114], [0.0380,0.0633], [0.6119,1.0105], [0.7932,1.2611] and [3.8281,6.3930], respectively.

human-related parameters. These results mean that reducing the number of direct contacts between susceptible sheep/humans and infectious sheep contributes to a lower peak number of infected individuals with two age groups (Fig. 7).

4.3. Tendency prediction

4.3.1. Tendency prediction without an extra-control measure

The Chinese government implemented two control measures of brucellosis in June 2024 (National Disease Control and Prevention Administration of the People's Republic of China., 2024) and September 2025 (The National People's Congress of People's Republic of China, (2025)). These two control measures respectively separate the 2024-2030 tendency predictions of cumulative infection scales into three phases in Figs. 8–11. The baseline measures, referred to as the recruitment rate, vaccination rate, culling rate, and direct transmission rates, are kept same in Table 1. The measures relating to these parameters are respectively denoted as single-control measure, double-control measure, triple-control measure and quadruple-control measure. The measures with the subscript *I* indicate that the corresponding parameter values are increased by 10% in June 2024, and by an extra 10% increase in September 2025, while those with the subscript *D* indicate that the corresponding parameter values are decreased by 10% in June 2024, and by an extra 10% decrease in September 2025.

The maximum and minimum of infection scales for Group G1 and Group G2 with four-type control measures are presented in Table 3 as of December 2030. We find that infection scales with single-control, double-control, and triple-control measures fall into the ranges with quadruple-control measure. More precisely, the maximum infection scale for Group G1 with quadruple-control measure is 4997, being 10 times of that with baseline measure, and the minimum infection scale is 249, being 0.49 times of that with baseline measure. The maximum infection scale for Group G2 with quadruple-control measure is 97,107, being 11 times of that with baseline measure, and the minimum infection scale is 3209, being 0.38

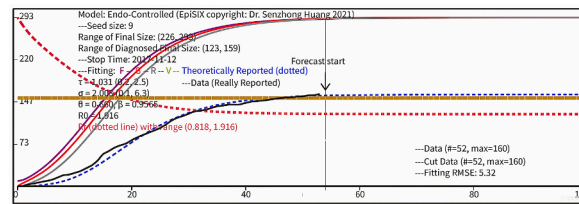


Fig. 4. EpiSIX fitting with Jinzhou brucellosis surveillance data in 2017, with basic reproduction number, 1.92; the range of final size of diagnosed individuals, [123, 159]; the percentage of Root Mean Square Error (RMSE), 3.3%; the number of weeks from beginning week of Jinzhou brucellosis epidemics to the week of effective reproduction number below one, 20 (Table 2).

Table 2

Main parameters and their values of Jinzhou brucellosis epidemics from 2017 to 2023 by EpiSIX.

Period (Start-End)	τ^a	σ^b	θ^c	β^d	R_o^e	RFSD ^f	RFS ^g	RMSE(%)	NW ^h
2017-01–2017-12	1.031	2.003	0.660	0.9565	1.92	[123, 159]	[226, 293]	3.3	20
2018-01–2018-12	1.002	1.975	0.710	0.9689	1.91	[153, 192]	[296, 371]	3.2	23
2019-01–2019-12	1.024	2.000	0.600	0.9520	1.90	[134, 182]	[270, 367]	2.6	30
2020-01–2020-12	1.120	2.514	0.646	0.9089	2.29	[282, 304]	[489, 528]	2.2	42
2021-01–2021-12	1.010	2.810	0.720	0.9564	2.69	[516, 714]	[891, 1232]	5.2	-
2022-01–2022-12	1.053	2.630	0.745	0.9500	2.50	[374, 400]	[603, 645]	4.1	31
2023-01–2023-12	1.003	2.391	0.733	0.9407	2.26	[228, 264]	[408, 473]	2.3	37

^a τ is the average incubation period.

^b σ is the average infectious period.

^c θ is the proportion of atypical patients.

^d β is the proportion of susceptible individuals.

^e R_0 is calculated by $R_0 = \beta\sigma$ in (Bai et al., 2023; Huang, 2008; Sun et al., 2023).

^f RFSD is the range of final size of diagnosed individuals.

^g RFS is the range of final size of diagnosed and undiagnosed individuals.

^h NW is the number of weeks from the beginning week of Jinzhou brucellosis epidemics to the week of effective reproduction number below one.

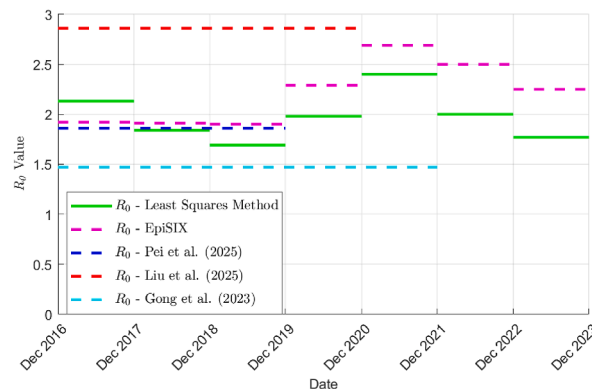


Fig. 5. Cross-validation is performed on basic reproduction numbers. The basic reproduction numbers in years by EpiSIX are derived from Table 2. The basic reproduction numbers by the least squares method for 2017, 2019, and 2023 are directly obtained from Fig. 3, and those for 2018, 2020, 2021, and 2022 are calculated by $[4 \times R_0^1 + 8 \times R_0^2]/12$, $[1 \times R_0^2 + 11 \times R_0^3]/12$, $[1 \times R_0^3 + 11 \times R_0^4]/12$, and $[4 \times R_0^4 + 8 \times R_0^5]/12$, where P_i and R_0^i are the i -th phase and the corresponding basic reproduction number with $R_0^1 = 2.13$, $R_0^2 = 1.69$, $R_0^3 = 2.00$, $R_0^4 = 2.44$, and $R_0^5 = 1.77$.

times of that with baseline measure. Furthermore, the maximum and minimum of basic reproduction numbers of the total population are estimated as 3.02 and 1.04 respectively. These results with quadruple-control measure indicate that the infection scales and the corresponding basic reproduction numbers describe the extreme impacts of brucellosis transmission risks, compared with other three-type control measures.

4.3.2. Tendency prediction with an extra-control measure

We assume that an extra-control measure of brucellosis is implemented by the Chinese government in December 2026. With two control measures introduced in June 2024 and September 2025 respectively, these three measures separate the 2024–2030 tendency predictions of cumulative infection scales into four phases in Figs. 12–15. The brucellosis infection scales with baseline measure are consistent with the ones in Figs. 8–11. The measures with the subscript I indicate that the

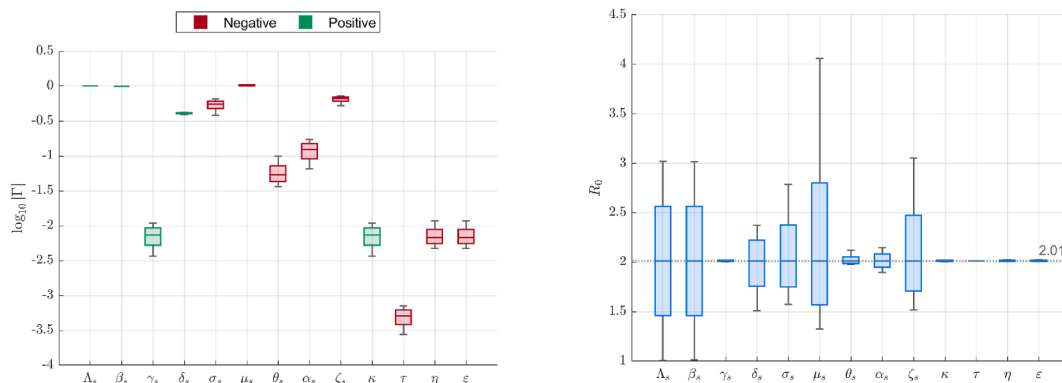


Fig. 6. The local sensitivity analysis of basic reproduction number with respect to each parameter. The sensitivity index $\Gamma = \Gamma(P) := \frac{\partial R_0}{\partial P} \frac{P}{R_0}$ and the impact magnitude $M := \log_{10}|\Gamma|$ of R_0 with respect to P are provided on left and right panels, where P is the parameter.

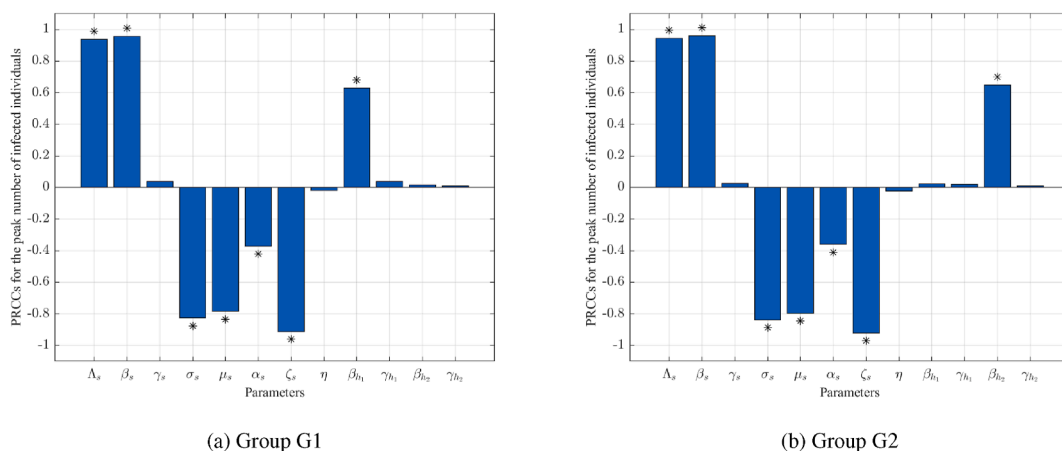


Fig. 7. The global sensitivity analysis of main parameters to the peak of Group G1 and Group G2 with sample size 3000. (*) denotes that the PRCCs are significant (p -value < 0.01).

corresponding parameter values are increased by 10% in June 2024, by an extra 10% in September 2025, and by a further extra 10% in December 2026, while those with the subscript D indicate that the corresponding parameter values are decreased by 10% in June 2024, by an extra 10% in September 2025, and by a further extra 10% in December 2026.

With the implementation of an extra-control measure, the maximum and minimum of infection scales for Group G1 and Group G2 with four-type control measures are presented in Table 3 as of December 2030. We find that infection scales with single-control, double-control, and triple-control measures also fall into the ranges with quadruple-control measure. More precisely, the maximum infection scale for Group G1 with quadruple-control measure is 12,176, being 24 times of that with baseline measure, and the minimum infection scale is 238, being 0.47 times of that with baseline measure. The maximum infection scale for Group G2 with quadruple-control measure is 232,531, being 28 times of that with baseline measure, and the minimum infection scale is 2988, being 0.35 times of that with baseline measure. Furthermore, the maximum and minimum of basic reproduction numbers of the total population are estimated as 3.67 and 0.90 respectively. The tendency predictions with an extra-control measure significantly impact on brucellosis infection scales than those without an extra-control measure. This means that an extra 10% increase in infection risk makes brucellosis infection scales with an extra-control measure sharp, exerting a potential and substantial effect on the overall scale of brucellosis infections (Figs. 11 and 15).

5. Discussion and conclusion

In this study, we develop a cross-species epidemic model for the description of Jinzhou brucellosis transmission, in which a sheep-brucella model and an age-structured human population model are incorporated with bilinear incidences. The bilinear incidence of sheep-brucella model relates to sheep population and brucella in the environment, and that of age-structured human population model relates to sheep population, brucella in the environment and human population.

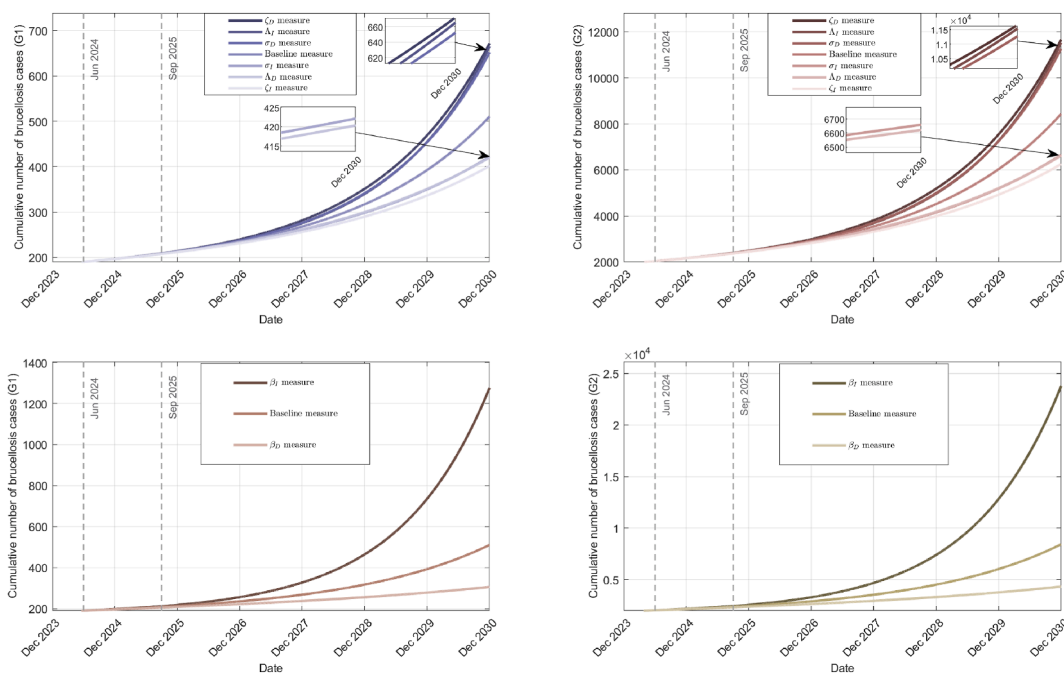


Fig. 8. The 2024–2030 tendency predictions of Jinzhou brucellosis epidemics with single-control measure.

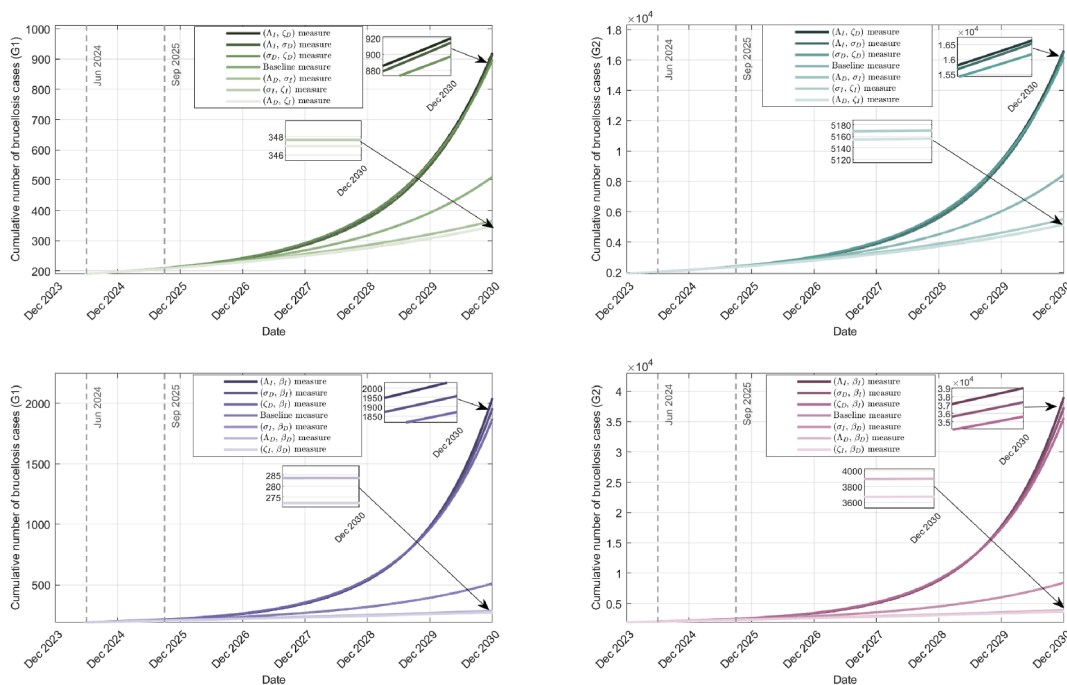


Fig. 9. The 2024–2030 tendency predictions of Jinzhou brucellosis epidemics with double-control measure.

Theoretically, we obtain basic reproduction number (i.e., R_0) of the cross-species brucellosis model in (3) via the next-generation matrix method. Meanwhile, the disease-free equilibrium point and the endemic equilibrium point are computed, and their stabilities are proved with the conditions in terms of basic reproduction number. Specifically, when $R_0 < 1$ is valid, the disease-free equilibrium point is locally asymptotically stable by Routh-Hurwitz Criterion in Theorem 3.1, and is further globally asymptotically stable by LaSalle's Invariance Principle in Theorem 3.2. When $R_0 > 1$ holds, the

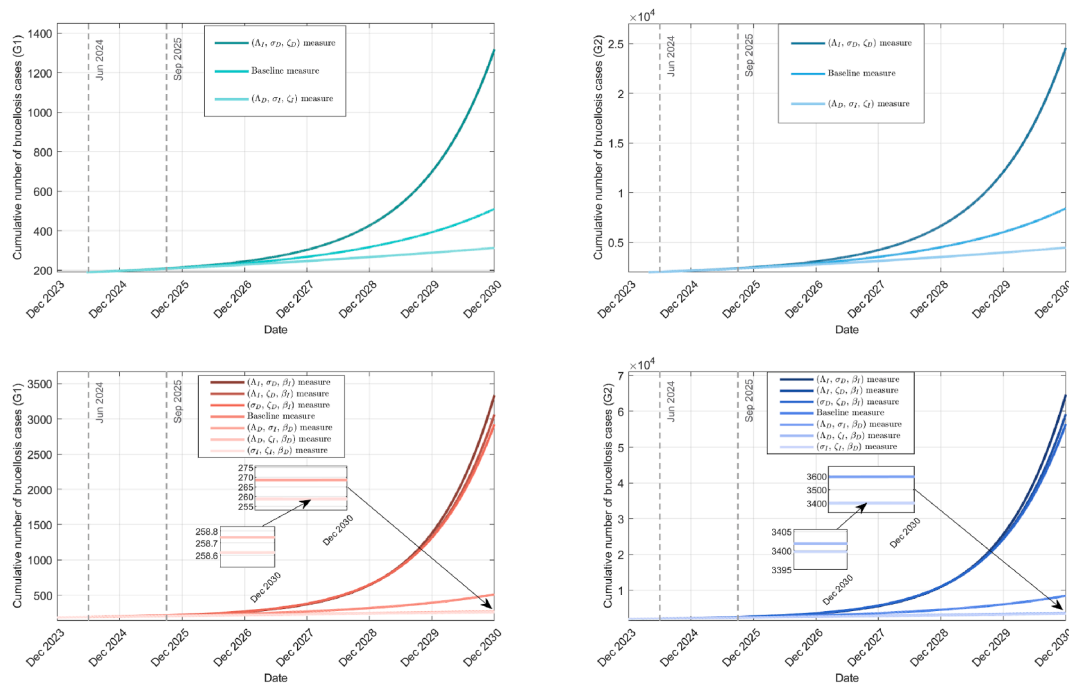


Fig. 10. The 2024–2030 tendency predictions of Jinzhou brucellosis epidemics with triple-control measure.

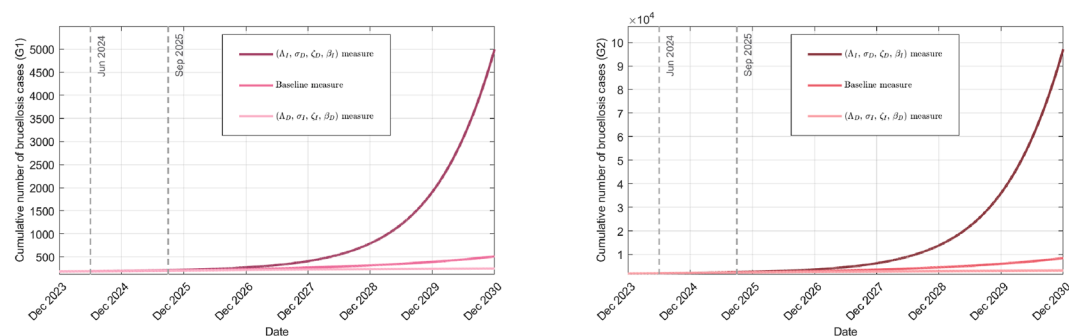


Fig. 11. The 2024–2030 tendency predictions of Jinzhou brucellosis epidemics with quadruple-control measure.

endemic equilibrium point is proved to be globally asymptotically stable by LaSalle's Invariance Principle again in [Theorem 3.3](#).

Numerically, we focus on the evolution of basic reproduction number for Jinzhou brucellosis epidemics. First of all, we collect the 2017–2023 monthly brucellosis surveillance data from Jinzhou CDC, and also collect two control measures and two livestock-related measures ([Fig. 2](#)), of which the implementation of two livestock-related measures triggers the promotion of livestock farming scale, further enhances the brucellosis infection scale in Jinzhou surveillance data. Next, we perform the optimal fitting of the cross-species brucellosis model by the least squares method, with one-month awareness delay and the main parameter values from [Table 1](#), and derive that basic reproduction number ranges from 1.69 to 2.40 ([Fig. 3](#)). Again, we use EpiSIX to fit the yearly cumulative brucellosis infection scales ([Fig. 4](#)) with the main parameter values from [Table 2](#), and also derive that basic reproduction number ranges from 1.90 to 2.60. The comparisons in [Fig. 5](#) reveal that basic reproduction numbers by two methods are close to those in ([Gong et al., 2023](#); [Liu et al., 2025](#); [Pei et al., 2025](#)). We further analyze the impacts of each parameter on basic reproduction number in [Fig. 6](#), which indicates that brucellosis transmission risks are effectively reduced by the decreases of direct transmission rate and recruitment rate, the increases of culling rate and vaccination rate.

Again, we numerically focus on the peak of infected individuals and the prediction results of infection scales of human population with two age groups. Firstly, we assess the impacts of main parameters on the peak of infected individuals in [Fig. 7](#), and find that the reduction of direct contact numbers between susceptible sheep/humans and infectious sheep

Table 3
The tendency predictions of Jinzhou brucellosis epidemics as of December 2030.

Measure	Parameter	Without an extra-control measure					With an extra-control measure				
		Group G1		Group G2		R_0^c	Group G1		Group G2		R_0^d
		IS ^a	SS ^b	IS ^a	SS ^b		IS ^a	SS ^b	IS ^a	SS ^b	
Baseline	Baseline	511	0	8431	0	1.77	511	0	8431	0	1.77
Single	ζ_D	672	161	11,660	3229	1.99	724	213	12,709	4278	2.07
Single	ζ_I	402	−109	6253	−2178	1.59	378	−133	5772	−2659	1.53
Single	β_I	1278	767	23,782	15,351	2.08	1946	1435	37,153	28,722	2.21
Single	β_D	306	−205	4335	−4096	1.49	282	−229	3871	−4560	1.41
Double	Λ_I, ζ_D	920	409	16,636	8205	2.35	1112	601	20,493	12,062	2.47
Double	Λ_D, ζ_I	347	−164	5156	−3275	1.33	324	−187	4687	−3744	1.38
Double	Λ_I, β_I	2042	1531	39,042	30,611	2.46	3910	3399	76,006	67,575	2.77
Double	ζ_I, β_D	272	−239	3672	−4759	1.34	254	−257	3296	−5135	1.22
Triple	$\Lambda_I, \sigma_D, \zeta_D$	1318	807	24,590	16,159	2.56	1817	1306	34,568	26,137	2.91
Triple	$\Lambda_D, \sigma_I, \zeta_I$	313	−198	4471	−3960	1.23	294	−217	4096	−4335	1.10
Triple	$\Lambda_I, \sigma_D, \beta_I$	3335	2824	64,639	56,208	2.68	7715	7204	149,467	141,036	3.11
Triple	$\sigma_I, \zeta_I, \beta_D$	259	−252	3400	−5031	1.24	244	−267	3108	−5323	1.11
Quadruple	$\Lambda_I, \sigma_D, \zeta_D, \beta_I$	4997	4486	97,107	88,676	3.02	12,176	11,665	232,531	224,100	3.67
Quadruple	$\Lambda_D, \sigma_I, \zeta_I, \beta_D$	249	−262	3209	−5222	1.04	238	−273	2988	−5443	0.90

^a IS is for infection scale, and the measures associating with the maximum and minimum infection scales for each subfigure in Figs. 8–15 are presented.
^b SS is for saved scale, which is calculated by taking the difference between the infection scale with specific measure and the infection scale with baseline measure. The saved scale is a positive (or, negative) number, which indicates that the measure is the one with maximum (or, minimum) infection scale in the corresponding subfigure.

^c R_0 is basic reproduction number, which is estimated by $(P_1^{wt} \cdot V_1^{wt} + P_2^{wt} \cdot V_2^{wt} + P_3^{wt} \cdot V_3^{wt}) / (P_1^{wt} + P_2^{wt} + P_3^{wt})$, where the periods of P_1^{wt} , P_2^{wt} and P_3^{wt} are respectively January 2024 to May 2024, June 2024 to August 2025, and September 2025 to December 2030. The values of V_i^{wt} are basic reproduction number for every period in (3) without an extra-control measure.

^d R_0 is basic reproduction number, which is estimated by $(P_1^w \cdot V_1^w + P_2^w \cdot V_2^w + P_3^w \cdot V_3^w + P_4^w \cdot V_4^w) / (P_1^w + P_2^w + P_3^w + P_4^w)$, where the periods of P_1^w , P_2^w , P_3^w and P_4^w are respectively January 2024 to May 2024, June 2024 to August 2025, September 2025 to November 2026, and December 2026 to December 2030. The values of V_i^w are basic reproduction number for every period in (3) with an extra-control measure.

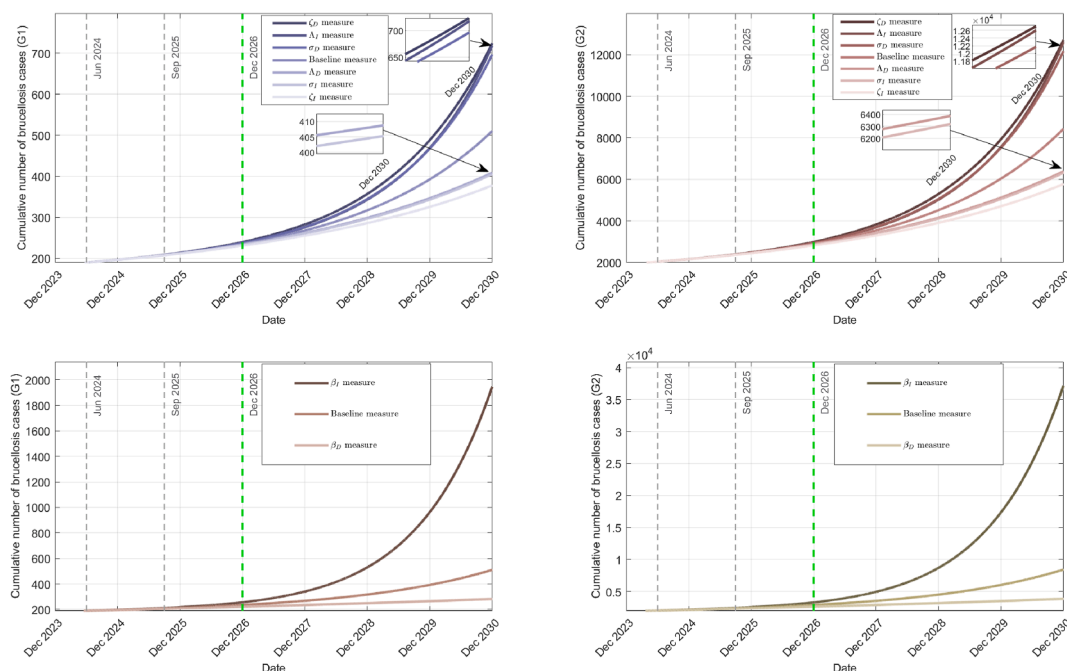


Fig. 12. The 2024–2030 tendency predictions of Jinzhou brucellosis epidemics with single- and extra-control measures.

declines the peak of infected individuals. Then, we discuss the 2024–2030 tendency predictions of brucellosis infection scales under two scenarios. When no extra-control measure is implemented, we find that brucellosis infection scales with quadruple-control measure describe the maximum and minimum transmission risks, compared to those with other three-type control measures (Figs. 8–11). When an extra-control measure is applied in December 2026, we further estimate the

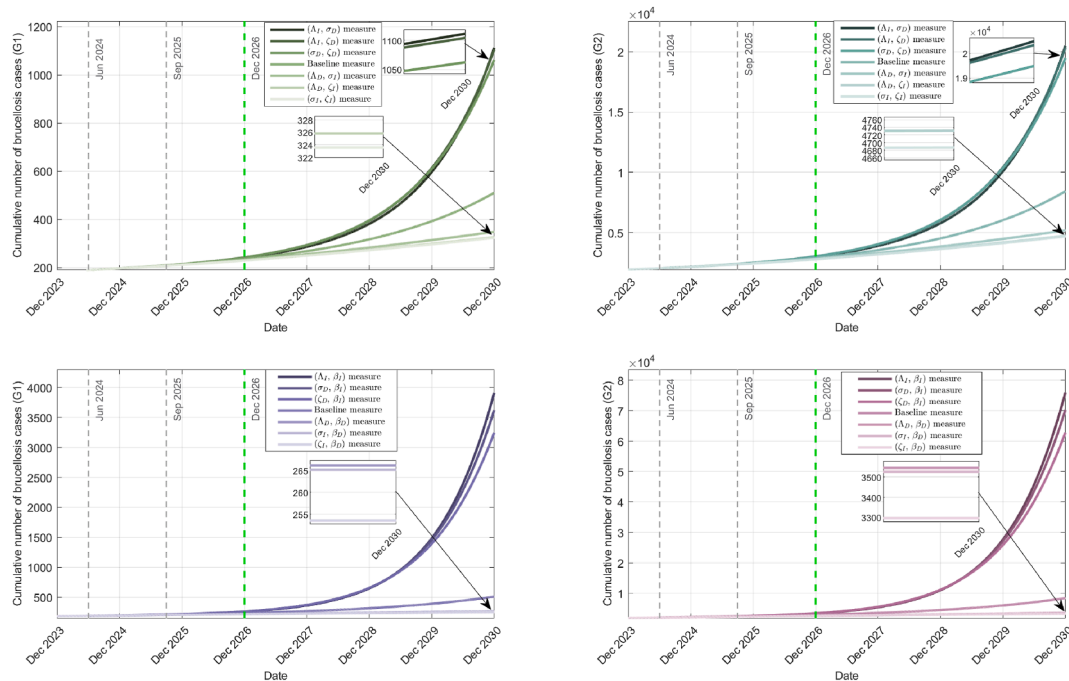


Fig. 13. The 2024–2030 tendency predictions of Jinzhou brucellosis epidemics with double- and extra-control measures.

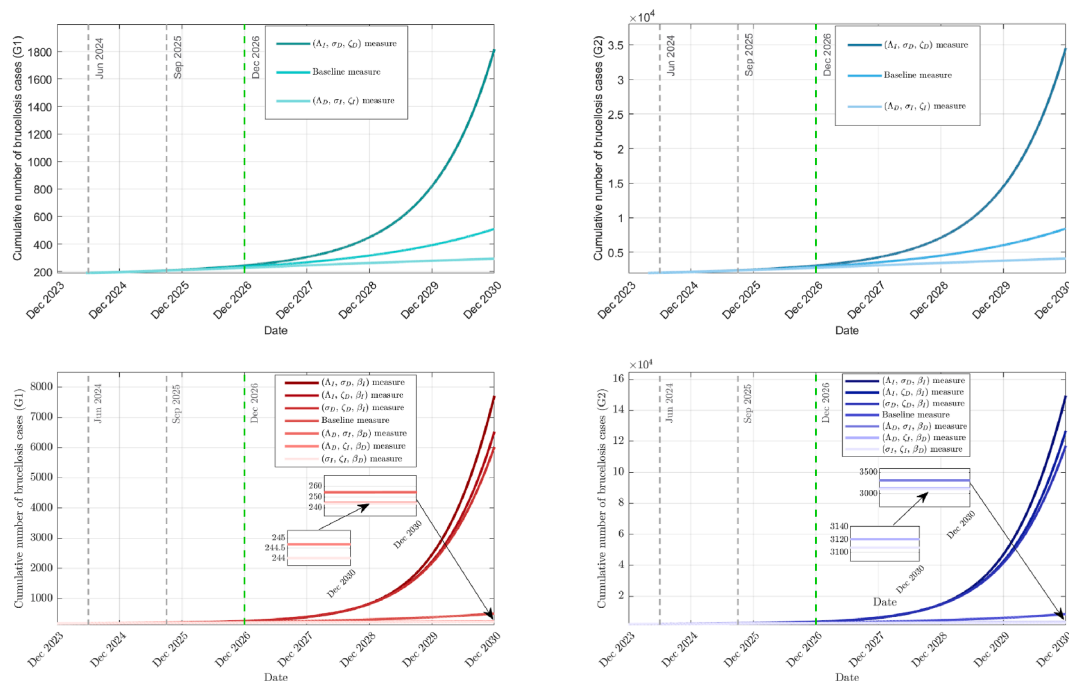


Fig. 14. The 2024–2030 tendency predictions of Jinzhou brucellosis epidemics with triple- and extra-control measures.

brucellosis infection scales with two age groups and four-type control measures (Figs. 12–15), and find that transmission risks are higher/lower than those without an extra-control measure (Table 3).

Compared with the cross-species brucellosis models proposed in (Gong et al., 2023; Liu et al., 2025; Pei et al., 2025), the first merit of this study is that we adopt age-structure of human population to characterize the differences of brucellosis transmission mechanisms during the spread of Jinzhou brucellosis epidemics. The estimations of basic reproduction

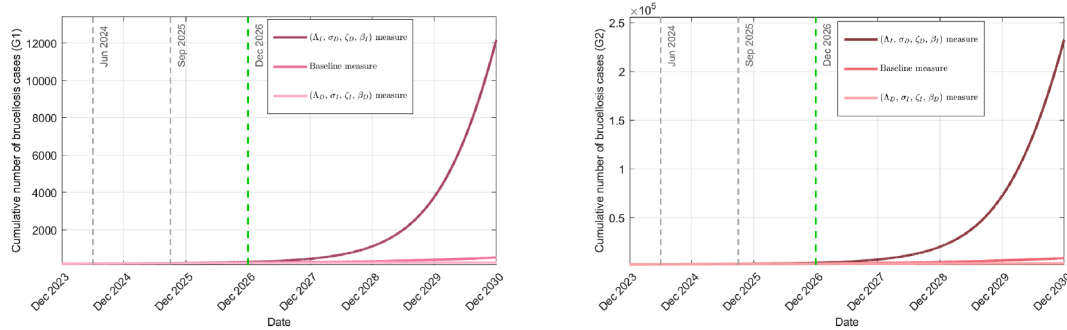


Fig. 15. The 2024–2030 tendency predictions of Jinzhou brucellosis epidemics with quadruple- and extra-control measures.

numbers of those three models are performed by the least squares method, so the second merit of this study is that we govern the least squares method and EpiSIX to fit Jinzhou brucellosis surveillance data for the accurate estimations of basic reproduction numbers. We also concern the effective reproduction number and the number of weeks that Jinzhou brucellosis epidemics are effectively controlled in years by EpiSIX, which becomes the third merit of this study. Meanwhile, we conduct tendency predictions under two scenarios, and analyze the specific infection scales with two age groups and four-type control measures in each scenario to depict the brucellosis transmission risks respectively, which becomes the fourth merit of this study.

More importantly, compared with studies (Gong et al., 2023; Li et al., 2019; Zheng et al., 2025) without human age-structure, the optimal fitting results of this study show that the direct transmission rate of Group G2 is significantly higher than that of Group G1 (Table 1). Meanwhile, the tendency predictions demonstrate that, when the key parameters ($\Lambda_s, \sigma_s, \zeta_s, \beta_s, \beta_h$) of two age groups are enhanced synchronously by the same proportion, the increase in the infection scale of Group G2 is much larger than that of Group G1. These results align with the epidemiological facts that Group G2 is the core population engaging in frequent contacts with infectious sheep during livestock production. Therefore, this study addresses the human age heterogeneity of transmission risk, avoiding the homogeneity of non age-structured models within the human populations. It further implies that control strategies and targeted interventions should prioritize Group G2, which represents the fifth merit of this work.

We propose several suggestions of brucellosis prevention and control, regarding the regions with similar sheep breeding patterns and geographical locations to Jinzhou, based on the above theoretical and numerical analysis. To reduce the recruitment rate of sheep population, it is recommended that the farmers adhere to self-breeding and self-rearing, and appropriately adjust sheep population to maintain its relative stable size. To reduce the direct transmission rates, the prevention and control measures are suggested to be implemented in two aspects. For declining sheep-to-sheep transmission, it is recommended that the farmers reduce grazing, optimize the layout of sheep houses, adopt zoned isolation breeding, and use dedicated tools. For declining sheep-to-human transmission, it is recommended that the relevant departments strengthen health education and publicity, popularize protection knowledge among relevant personnel and the general public through social media and other channels, conduct protection training and guidance, and call for annual brucellosis monitoring for relevant personnel, especially those from Group G2. To improve the culling rate of infectious sheep, it is recommended that relevant departments and the farmers implement daily inspections, promptly isolate sheep with suspected clinical symptoms and timely tests, cull positive sick sheep, and keep proper records with early detection and reporting. To increase the vaccination rate, it is recommended that the farmers and veterinarians standardize the immunization procedure of susceptible sheep, use disposable needles, regularly test antibodies and timely conduct booster vaccination. The more targeted measures from the local governments should be announced with the detailed guidance to reduce brucellosis transmission risks.

In future work, we plan to investigate the brucellosis epidemics from one or more of the following aspects. First, we will consider different incidences such as standard incidence and saturated incidence to explore their effects on the transmission dynamics of brucellosis (Zhang et al., 2014). Second, we will incorporate seasonal factors into the dynamical model to analyze the impact of seasonal variations on transmission risk (Ma et al., 2022). Third, we will extend the research perspective from macroscopic population-level mechanism to the microscopic cell-level mechanism underlying the interactions between pathogens and host cells, so as to provide new theoretical evidence for the prevention and control of brucellosis (Li et al., 2025). Fourth, we will introduce data-driven methods using neural ordinary differential equations to analyze local epidemics and achieve relatively accurate tendency predictions of brucellosis (Xiao et al., 2025).

CRediT authorship contribution statement

Shasha Chen: Writing – original draft, Visualization, Formal analysis, Data curation, Conceptualization. **Fengying Wei:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition. **Xinyan Zhang:** Writing

– review & editing, Data curation. **Hao Jin:** Writing – review & editing, Data curation. **Yan Zhao:** Writing – review & editing. **Shuang Zhang:** Writing – review & editing.

Ethics approval and consent to participate

The ethical approval and individual consents were exempted as the aggregated data were used in this study.

Data and material availability

The data that supported this article may be available upon reasonable request to the corresponding author.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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